Nuclear Technology Development and Economics
2025

Current Trends in the Supply and Utilisation of Medical Radioisotopes

OECD
BETTER POLICIES FOR BETTER LIVES

NEA
NUCLEAR ENERGY AGENCY

Nuclear Development

Current Trends in the Supply and Utilisation of Medical Radioisotopes

NEA No. 7743

NUCLEAR ENERGY AGENCY
ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

The OECD is a unique forum where the governments of 38 democracies work together to address the economic, social and environmental challenges of globalisation. The OECD is also at the forefront of efforts to understand and to help governments respond to new developments and concerns, such as corporate governance, the information economy and the challenges of an ageing population. The Organisation provides a setting where governments can compare policy experiences, seek answers to common problems, identify good practice and work to co-ordinate domestic and international policies.

The OECD member countries are: Australia, Austria, Belgium, Canada, Chile, Colombia, Costa Rica, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Korea, Latvia, Lithuania, Luxembourg, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, the Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Türkiye, the United Kingdom and the United States. The European Commission takes part in the work of the OECD.

OECD Publishing disseminates widely the results of the Organisation's statistics gathering and research on economic, social and environmental issues, as well as the conventions, guidelines and standards agreed by its members.

This work is published under the responsibility of the Secretary-General of the OECD. The opinions expressed and arguments employed herein do not necessarily reflect the official views of the member countries of the OECD or its Nuclear Energy Agency.

NUCLEAR ENERGY AGENCY

The OECD Nuclear Energy Agency (NEA) was established on 1 February 1958. Current NEA membership consists of 34 countries: Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Korea, Luxembourg, Mexico, the Netherlands, Norway, Poland, Portugal, Romania, Russia (suspended), the Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Türkiye, the United Kingdom and the United States. The European Commission and the International Atomic Energy Agency also take part in the work of the Agency.

The mission of the NEA is:

to assist its member countries in maintaining and further developing, through international co-operation, the scientific, technological and legal bases required for the safe, environmentally sound and economical use of nuclear energy for peaceful purposes.

to provide authoritative assessments and to forge common understandings on key issues as input to government decisions on nuclear energy policy and to broader OECD analyses in areas such as energy and the sustainable development of low-carbon economies.

Specific areas of NEA competence include the safety and regulation of nuclear activities, radioactive waste management, radiological protection, nuclear science, economic and technical analyses of the nuclear fuel cycle, nuclear law and liability, and public information. The NEA Data Bank provides nuclear data and computer program services for participating countries.

Please cite this publication as:

NEA (2025), *Current Trends in the Supply and Utilisation of Medical Radioisotopes,* OECD Publishing, Paris, https://doi.org/10.82155/x9kb-7j83.

Version 1.1, 21 October 2025

Foreword

In past decades, a cancer diagnosis was widely perceived as a death sentence. As medical diagnostic and therapeutic technology has advanced – in large part through nuclear technologies – this is now far less true. Cancers are detected more quickly and treated more effectively today than could have been imagined in the past.

The advanced nuclear medicine technologies that are facilitating this progress rely on a steady supply of medical radioisotopes, which has been a challenge to ensure in the past. The OECD Nuclear Energy Agency (NEA) has made this one of the points of focus of its activities since 2009, a period marked by significant shortages of molybdenum-99 (Mo-99) and its derivative product, technetium-99m (Tc-99m). The international community called on the NEA to review the supply challenges and find ways to prevent such crises. This led to the establishment of the NEA's High-Level Group on the Security of Supply of Medical Radioisotopes (HLG-MR).

Comprising experts from 18 countries, including non-NEA member countries, the Euratom Supply Agency (ESA) and the International Atomic Energy Agency (IAEA), the HLG-MR informed policy decisions to stabilise supplies. Additional organisations such as the European Observatory on the Supply of Medical Radioisotopes and Nuclear Medicine Europe (NMEU) have also contributed to mitigating risks to the security of supply.

The HLG-MR completed its work on the subject in 2018 following several years of relative stability. However, subsequent years, including the COVID-19 pandemic in 2020, revealed continued vulnerabilities in the supply of medical radioisotopes and the NEA re-engaged the question in 2023. Work in 2025-2026 will focus on assessing market demand growth potential, supply chain vulnerabilities (e.g. enrichment, irradiation, processing, treatment), and gap analysis of medical infrastructure considerations. As new medical radioisotopes for treating diseases become more widely available, this may open new commercial models that improve the market economics of Mo-99. The NEA will continue to actively engage with the topic of security of supply with policy recommendations and outage reserve capacity (ORC) models for the benefit of its member countries.

This report outlines the current landscape for the supply of medical radioisotopes and identifies decisions that will need to be made to accelerate and expand access to these life-saving technologies. It is based on work by the HLG-MR and the contributions of participants to its activities, including the International Workshops on the Security of Supply for Medical Radioisotopes, held in 2023 and 2024.

The first of these workshops evidenced continued concerns about the reliability of Mo-99 supplies and optimism about the emergence of a new generation of medical radioisotopes with therapeutic applications. The second edition of the event showcased innovations using radioisotopes such as lutetium-177 (Lu-177) and actinium-225 (Ac-225) in cancer treatment. However, it has yet to be proven how and when the supply will be able to meet demand.

Acknowledgements

The NEA expresses its gratitude to the governments of Japan and the United States for their continued support of the Agency's work on the supply of medical radioisotopes. Their engagement has been instrumental in advancing the Agency's efforts to improve resilience and sustainability in this critical domain.

Appreciation is also extended to the other NEA divisions that contributed to the development of this document: the Division of Nuclear Safety Technology and Regulation, the Division of Radioactive Waste Management and Decommissioning, and the Division of Radiological Protection and Human Aspects of Nuclear Safety. Their expert reviews and constructive feedback have enhanced the quality and scope of this work.

The NEA acknowledges the valuable input from radioisotope producers, who shared essential data on their future activities and strategies aimed at strengthening supply chain resilience. The NEA is equally grateful to pharmaceutical companies working in the field of therapeutic radioisotopes for their perspectives and information, which have helped to enrich the analysis presented in this report.

Anikitos Garofalakis of the NEA Division Nuclear Technology Development and Economics was the lead author of this report.

Table of contents

List of abbreviations and acronyms **7**

Executive summary **9**

Chapter 1. Introduction **11**

1.1 Overview of medical radioisotopes in nuclear medicine and the emergence of radioligand therapy (RLT) 11
1.2 Impact on global health and the need for security of supply 15

Chapter 2. Supply of medical radioisotopes in nuclear medicine: Challenges and perspectives **17**

2.1 Current situation and challenges of medical radioisotopes production 17
2.2 Supply of lutetium-177 for radioligand therapy 20
2.3 Resilience strategies for Mo-99 21
2.4 Resilience strategies for Lu-177 26
2.5 Waste management, transportation and radiological protection 26

Chapter 3. Radioligand therapy for cancer treatment **29**

3.1 Significance of RLT for cancer treatment: An innovative therapeutic modality 29
3.2 Approved drugs and clinical trials: current progress and prospects 30
3.3 Theranostics: Benefits from the integration of diagnostics and therapeutics 32
3.4 Description of a radioligand therapy session: Clinical workflow and patient experience 32
3.5 Infrastructure needs for enhancing diagnostic and therapeutic capabilities 34

Chapter 4. Key findings and areas for possible future work **37**

4.1 Supply and demand forecasting 37
4.2 Expert group at the NEA 37
4.3 International market transparency and monitoring of supply 38
4.4 National policy frameworks 39
4.5 Regulatory efficiency 39
4.6 Enabling innovation 40

Chapter 5. Conclusions **41**

5.1 Summary of key challenges and opportunities 41
5.2 Ensuring the future of medical radioisotope supply 42
5.3 The role of policy in strengthening the radioisotope supply chain 42
5.4 Next steps 42

Glossary **45**

References **47**

List of tables

Table 1 Examples of important radiopharmaceuticals for diagnosis and therapy 15
Table 2 Research reactors with significant Mo-99 production capacity 17
Table 3 Summary of major novel medical radioisotope production projects and technologies 21

List of figures

Figure 1 The molybdenum-99 and technetium-99m supply chain 11
Figure 2 Radioactive compound linked to a ligand targets protein and causes cancer cell death 14
Figure 3 Map of major research reactors and processing facilities for Mo-99 production 19
Figure 4 Step-by-step process of RLT, from drug preparation to targeted cancer cell destruction 30
Figure 5 Clinical trial results for RLT based on Lu-177 31
Figure 6 RLT workflow 33

List of abbreviations and acronyms

ALARA	As low as reasonably achievable
CANDU	Canada Deuterium Uranium
CMIE	Canadian Medical Isotope Ecosystem
CPDC	Centre for Probe Development and Commercialization (Canada)
DRG	Diagnosis-related group
ESA	Euratom Supply Agency
FCR	Full-cost recovery
GMP	Good manufacturing practice
HEU	Highly-enriched uranium
HFR	High Flux Reactor
IAEA	International Atomic Energy Agency
IPS	Isotope Production System
IRE	Institute for Radioelements (Belgium)
ITM	Isotope Technologies Munich
JHR	Jules Horowitz Reactor (France)
LEU	Low-enriched uranium
LMIC	Low and middle-income countries
NCA	Non-carrier-added
NEA	Nuclear Energy Agency
NMEU	Nuclear Medicine Europe
NRC	Nuclear Regulatory Commission (United States)
NRU	National Research Universal reactor
OPG	Ontario Power Generation
ORC	Outage reserve capacity
PPE	Personal protective equipment

PRRT	Peptide receptor radionuclide therapy
PSMA	Prostate-specific membrane antigen
R&D	Research and development
RLT	Radioligand therapy
SON	Saugeen Ojibway Nation
SPECT	Single Photon Emission Computed Tomography
TDS	Target Delivery System
U.S. FDA	United States Food and Drug Administration

Executive summary

Cancer is one of the leading causes of death worldwide, with an estimated 19.3 million new cases and 10 million fatalities reported in 2020 (Sung et al., 2021). Among the approaches taken by physicians to diagnose and treat various cancers, nuclear medicine techniques have proven to be very effective in terms of both the quality of life of patients and the outcomes realised as a result of these interventions. Medical radioisotopes, materials resulting from nuclear reactions that are a key component of nuclear medicine, are key to most nuclear medicine procedures. The supply of medical radioisotopes, however, has not always been stable, with notable shortages in 2009 and after the COVID-19 pandemic. This has hindered progress in this otherwise promising area of medicine.

The production of medical radioisotopes relies on a fragile supply chain that revolves around a limited number of facilities around the world. Maintaining this supply chain has, at times, required direct intervention by governments, and challenges persist, including: the ageing of the reactor infrastructure used to produce technetium-99m (Tc-99m) from molybdenum-99 (Mo-99); the technical difficulties of using low-enriched uranium (LEU) in place of highly enriched uranium (HEU); and high production costs. This report focuses on how to improve the sector's resilience and sustainability, including by modernising infrastructure and diversifying production methods.

This document also examines the potential for adopting radioligand therapy (RLT), a highly promising innovation in treating cancer. The challenges to RLT adoption include a reliance on radioisotopes that have half-lives and radiation characteristics different from those to which the supply chain is accustomed. As a result, it will be necessary to adapt and expand existing regulatory frameworks and to create specialised infrastructure, such as treatment facilities with advanced imaging and radiopharmacy capabilities, and a (re)trained workforce if RLT is to be delivered at scale. Investing in training for nuclear medicine professionals and strengthening international collaboration on regulatory alignment could aid this process, particularly in low-resource settings. Meanwhile, creating reimbursement mechanisms will facilitate access to RLT and ensure its long-term sustainability.

This report concludes by outlining six areas for possible future work in NEA member countries that would aim to improve global health outcomes through the timely adoption of RLT and innovations in medical radioisotope delivery. The potential next steps in this area are:

- forecasting supply and demand through periodic assessments by the NEA, including for radioisotopes needed for novel procedures;
- establishing a new Expert Group at the NEA to monitor production and demand trends, assess the pipeline of innovative radioisotopes, and provide follow-up on policy recommendations;
- enhancing international co-operation on market transparency and the monitoring of supply;
- establishing national policy frameworks, with relevant ministries, regulators, and industry stakeholders improving co-ordination on medical radioisotope production and supply;

- improving regulatory efficiency through international collaboration and sharing of experience and best practices; and
- enabling innovation through, among other things, specialised infrastructure and logistics, efficient regulatory pathways, and workforce training.

Chapter 1. Introduction

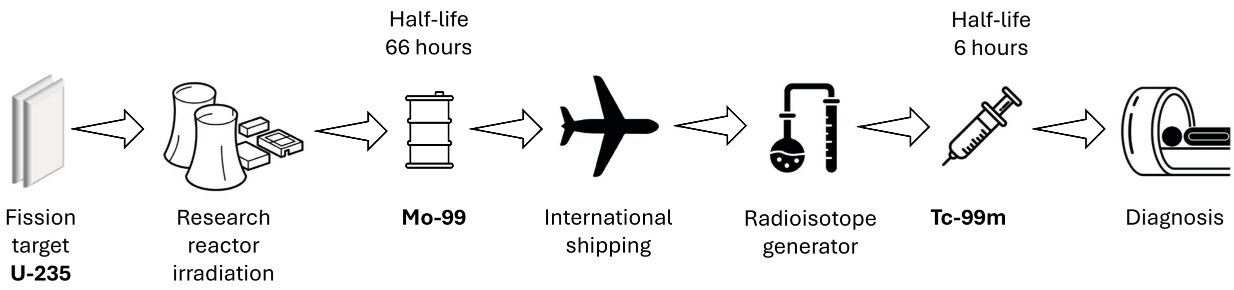
1.1 Overview of medical radioisotopes in nuclear medicine and the emergence of radioligand therapy (RLT)

Medical radioisotopes are the cornerstone of nuclear medicine. Since their discovery, their unique properties have been harnessed to analyse body functions and deliver life-saving insights. These radioisotopes enable the production of early, precise and detailed 2D or 3D internal images, helping physicians plan treatments effectively. The non-invasive nature of this procedure reduces patient discomfort while offering critical diagnostic information.

Technetium-99m (Tc-99m) accounts for most nuclear medicine procedures globally, with estimates suggesting usage rates in 80-85% of cases. Tc-99m is employed in over 30 million procedures annually, making it the backbone of nuclear medicine diagnostics (NEA, 2018). Once introduced into the body, Tc-99m accumulates in targeted tissues and emits gamma rays, which are captured by gamma cameras or Single Photon Emission Computed Tomography (SPECT) systems to deliver detailed images. These are used in various applications, including cardiovascular imaging, cancer detection, infection localisation, bone scans and thyroid evaluations.

A key feature of Tc-99m that contributes to its widespread utilisation is its short half-life of only 6 hours. While this requires strict protocols for handling, transportation, and waste management, the rapid decay significantly reduces radiation exposure to the patient and minimises environmental risks. Tc-99m is typically produced using generators loaded with molybdenum-99 (Mo-99), its parent isotope. Its longer half-life of 66 hours allows Mo-99 to be shipped worldwide, allowing Tc-99m to be generated on-site as needed for just-in-time production and shipment to radiopharmacies (see Figure 1).

Figure 1 The molybdenum-99 and technetium-99m supply chain



The supply chain for Mo-99 faces significant vulnerabilities. Only a limited number of reactors globally produce Mo-99 at scale. This global dependence on a limited number of reactors for Mo-99 production has resulted from a combination of historical, economic, regulatory and technical factors. Initially, Mo-99 production was concentrated in government-funded research reactors, many of which were built in the 1950s–1970s to support nuclear research, with isotope production as a secondary function. Over time, as demand for Mo-99 grew, few new reactors were built, leaving the existing facilities to serve the market.

The 2008-2009 Mo-99 supply crisis, triggered by the unplanned shutdown of Canada's National Research Universal (NRU) reactor, a full-scale Canada Deuterium Uranium (CANDU) research reactor that operated from 1957 to 2018, and which produced a large majority of the Mo-99 supply, led to widespread cancellations of medical appointments, jeopardising patient outcomes globally. This event revealed the fragility of the supply chain and led to international efforts to improve the resilience. This prompted NEA member countries to establish the High-Level Group on the Security of Supply of Medical Radioisotopes (HLG-MR) in 2009, co-ordinated by the NEA Secretariat, to address these challenges through policy recommendations and collaborative frameworks to prevent future crises. Work under the HLG-MR concluded in 2018 following a period of extended stability in the security of supply.

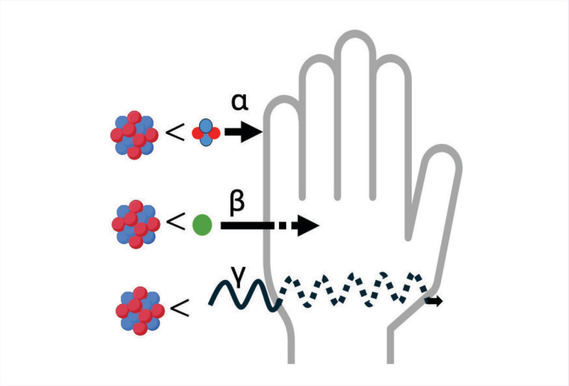
Therapeutic radioisotopes

Traditionally, medical radioisotopes have been used as diagnostic tools. Since 2010 innovative applications in their therapeutic use have opened up new possibilities for treating tumours.

Therapeutic use of medical radioisotopes is not a recent development. As early as the early 1940s, medical use of iodine-131 (I-131) to treat disease began. Upon injection, I-131 accumulates in the thyroid gland where it emits beta radiation, destroying the diseased cells associated with hyperthyroidism and thyroid cancer. More recently, advancements in alpha-emitting radioisotopes have shown promise for therapeutic applications. As an example, Xofigo (radium-223 dichloride), which was approved by the United States Food and Drug Administration (U.S. FDA) in 2013, represents an important advancement in the treatment of bone metastases from metastatic castration-resistant prostate cancer. By mimicking calcium, radium-223 targets areas of high bone turnover, emitting alpha particles to destroy cancerous cells while sparing healthy tissue. However, Xofigo's use remains relatively limited compared to I-131 due to a narrower range of usages and significant cost factors.

Box 1: Alpha, beta and gamma ray emitters explained

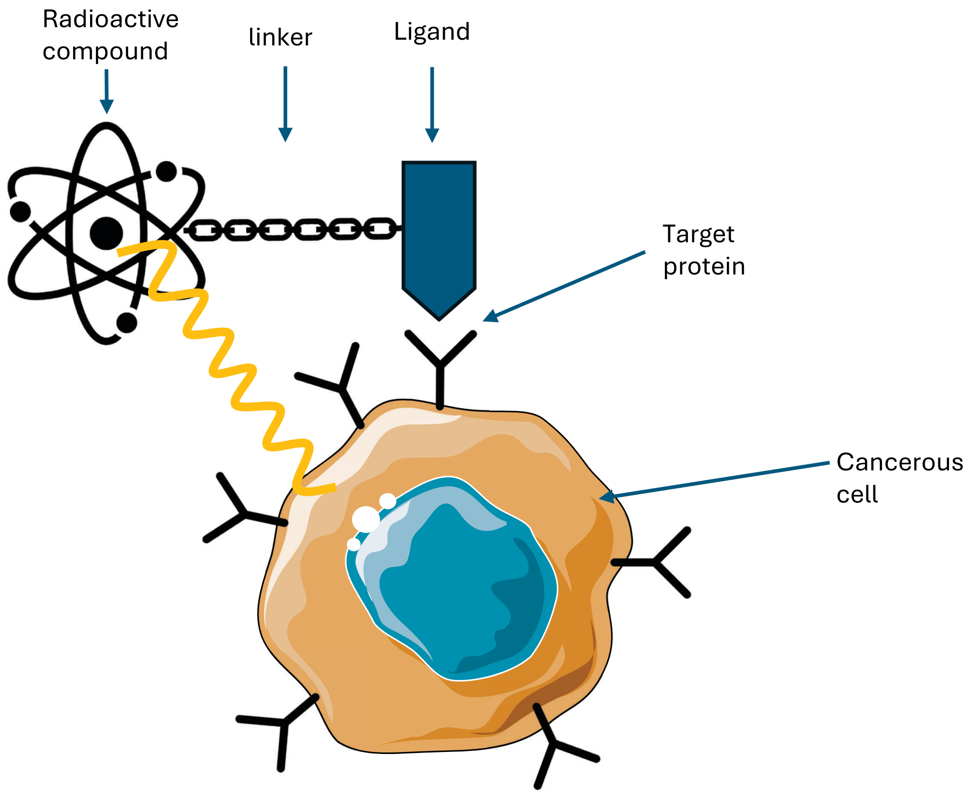
Alpha emitters are highly energetic radionuclides that emit alpha particles, consisting of two protons and two neutrons, tightly bound together. Alpha particles, because they are highly ionising, are unable to penetrate very far through matter and are brought to rest by a few centimetres of air or less than a tenth of a millimetre of biological tissue (ARPANSA, n.d.a). Alpha radiation causes double-strand DNA breaks in cancer cells, leading to cell destruction while sparing healthy tissue. Examples of alpha emitters used in medicine include Actinium-225 (Ac-225) and Astatine-211 (At-211). **Beta emitters** are radionuclides that decay by emitting beta particles during a form of radioactive decay called beta decay. Beta particles are high-energy, high-speed electrons (β-) or positrons (β+). Beta particles, being less ionising than alpha particles, can travel through millimetres of skin or tissue (ARPANSA, n.d.b). In the case of beta emitters, the main mechanism of DNA damage is indirect ionisation by free radicals. This makes them suitable for targeted radionuclide therapy, most especially for small or diffuse tumours. Examples of beta emitters include Lutetium-177 (Lu-177), Yttrium-90 (Y-90), and I-131. **Gamma-ray emitters** are radionuclides that decay by emitting high-energy electromagnetic radiation (gamma rays, γ). Contrary to alpha or beta particles, gamma rays have high penetration capacity, allowing them to travel many centimetres inside tissue. This property makes them ideal for diagnostic imaging in nuclear medicine, where they can exit the body and be detected by external gamma cameras, SPECT or PET scanners Examples of gamma emitters include Tc-99m, Iodine-123 (I-123), and Fluorine-18 (F-18).



Despite significant progress made in oncology in treatments like chemotherapy, hormone therapy, radiotherapy and immunotherapy, there are still gaps in the needs of patients with rare, aggressive or late-stage cancers. Many existing therapies are insufficient to treat these cases, leaving a pressing need for innovative approaches to improve survival rates and quality of life.

Targeting cancers in tissues that lack a natural affinity for medical radioisotopes (like I-131 and radium-223 for thyroid and bone respectively) has become possible thanks to advances in molecular biology, which have enabled the development of ligands – molecules designed to bind specifically to proteins or receptors on cancer cells (see Figure 2). Attaching a radioisotope to a ligand makes it possible to deliver radiation precisely to cancer cells, with the ligand acting as a "biological GPS". The technique of combining the precision of molecular targeting with the therapeutic power of radioisotopes, is commonly referred to as radioligand therapy (RLT).

Figure 2 Radioactive compound linked to a ligand targets protein and causes cancer cell death



Approvals of lutetium-177 DOTATATE (Lutathera®) in 2018 for gastroenteropancreatic neuroendocrine tumours (GEP-NETs) and lutetium-177 vipivotide tetraxetan (Pluvicto®) in 2022 in the United States for metastatic castration-resistant prostate cancer (mCRPC) highlight the growing clinical adoption of RLT. Both pharmaceuticals employ the beta-emitter lutetium-177 (Lu-177), which has become a cornerstone of RLT due to its ability to efficiently destroy tumours while sparing healthy tissue. GEP-NETs are characterised by high expression of somatostatin receptor subtype 2 (SSTR2), making them suitable targets for lutetium-177-based therapy. This approach utilises oxodotreotide (DOTATATE), a ligand that specifically binds to SSTR2, enabling targeted radiation delivery. Similarly, lutetium-177 vipivotide tetraxetan is designed to treat metastatic castration-resistant prostate cancer by employing vipivotide tetraxetan (PSMA-617), which selectively binds to prostate-specific membrane antigen (PSMA) for precise radiopharmaceutical therapy. Efforts are also underway to develop therapies using other beta and alpha emitters (see Box 1), such as actinium-225 (Ac-225), lead-212 (Pb-212), and astatine-211 (At-211), which deliver higher energy over a shorter range, making them ideal for treating micrometastases (Jang et al., 2023).

Table 1 below summarises the approved radiopharmaceuticals so far, along with key details about diagnostic and/or therapeutic applications.

Table 1 Examples of important radiopharmaceuticals for diagnosis and therapy

Radio-pharmaceutical	Abbreviation/ Commercial name	Radio-isotope used	Emitter	Half-life	Diagnostic/therapeutic	Medical applications	Key producers	Patients benefited annually
Molybdenum-99	Mo-99	Moly-bdenum-99	Beta	~66 hours	Precursor for diagnosis	Parent isotope for Tc-99m generators, used in nuclear imaging	NTP (South Africa), ANSTO (Australia), Curium (Netherlands), IRE (Belgium)	Used to generate Tc-99 for millions of patients annually (NEA, 2018)
Technetium-99m	Tc-99m	Techneti-um-99m	Gamma	~6 hours	Diagnostic imaging	Used in SPECT scans for cardiac, bone, lung, kidney, and cancer imaging	Generated on-site from Mo-99; supplied through Mo-99 producers	30-40 million procedures per year worldwide (NEA, 2018)
Sodium iodide I-131	NaI-131	Iodine-131	Beta & Gamma	~8 days	Therapeutic & diagnostic	Thyroid cancer, hyper-thyroidism	INIS, Jubilant DraxImage, ANSTO, Nordion, PDR Radiopharma	Hundreds of thousands annually[1]
Lutetium-177 dotatate	Lutathera	Lutetium-177	Beta	~6.7 days	Therapeutic	Neuroendocrine tumours (NETs)	Novartis	Thousands per year (Czernin and Calais, 2022)
Lutetium-177 vipivotide tetraxetan	Pluvicto	Lutetium-177	Beta	~6.7 days	Therapeutic	Metastatic castration-resistant prostate cancer (mCRPC)	Novartis	Thousands per year (Farolfi et al., 2024)
Radium-223 dichloride	Xofigo	Radium-223	Alpha	~11.4 days	Therapeutic	Bone metastases from prostate cancer	Bayer	Thousands per year (Deshayes et al., 2017)

1.2 Impact on global health and the need for security of supply

Medical radioisotopes have profoundly influenced health care around the world. Where they are available, they provide critical diagnostic capabilities that allow early detection, precise disease staging, and improved treatment planning. However, worldwide access remains very unevenly distributed (IAEA, 2025).

The development of RLT has enabled the use of radioisotopes for therapeutic purposes, offering an innovative way of treating cancer in the context of precision medicine. This technology represents another tool in the effort to combat cancer and is expected to provide more efficient treatment compared to conventional treatments.

The synergy between diagnostic imaging and therapeutic intervention is exemplified in theranostics, which integrates both processes into a seamless framework. Diagnostic imaging is essential for planning therapy by confirming target expression and assessing patient eligibility for RLT. For instance, in the treatment of neuroendocrine tumours (which often present as multiple lesions), PET/CT imaging with Gallium-68 DOTATATE helps detect tumours that express somatostatin receptors. DOTATATE is a ligand that specifically binds to these receptors, permitting targeted imaging when labelled with Gallium-68. This process enables precise visualisation of tumours and helps identify

1 Estimation based on epidemiological data on hyperthyroidism and thyroid cancer.

patients who are suitable candidates for Lu-177 DOTATATE therapy. During therapy, the Lu-177 attached to the DOTATATE ligand delivers radiation directly to cancerous cells, selectively destroying them while minimising damage to healthy tissue. This example demonstrates how diagnostic and therapeutic radioisotopes work in a combined manner to deliver treatment tailored to a patient's specific molecular profile, maximising therapeutic efficacy while minimising unnecessary radiological exposure.

As outlined earlier in this report, the supply chain of Mo-99 for diagnostics is particularly vulnerable. In the case of RLT treatment, ensuring a resilient supply chain for Lu-177 as well as other radioisotopes is equally critical due to their specific characteristics.

The just-in-time nature of medical radioisotopes that stem from their short half-lives, prohibit stockpiling and thus require precise implementation and co-ordination of the manufacturing and delivery processes. There is also a need for rigorous logistical protocol due to the intrinsic radioactivity of these isotopes. These complex challenges disproportionately affect low and middle-income countries (LMICs), and as a result, there is limited access to vital diagnostic and therapeutic tools for their health care systems.

Additionally, the potential of RLT to ameliorate cancer treatment can only be realised if health care systems are equipped to deliver this therapy. Specialised infrastructure is thus crucial for adopting RLT and addressing issues such as waste management and radiological protection. Additionally, raising awareness of RLT among nuclear medicine professionals can enable its implementation and facilitate the creation of reimbursement mechanisms to tackle accessibility and sustainability issues.

The synergy between diagnostic imaging and therapeutic action highlights the need for co-ordinated policy actions. Efforts to modernise medical radioisotopes production facilities, diversify processing centres, and address supply chain vulnerabilities will need to continue if global access to these innovative tools is to be facilitated. In parallel, the rapid rise of RLT and theranostics, an approach that combines targeted diagnostics and radionuclide therapy using the same ligand, necessitates strategic investments to build infrastructure, train professionals and secure a sustainable market. By addressing these challenges, policymakers can provide the enabling conditions for increased access and timely adoption of these technologies, thereby ultimately improving the landscape of nuclear medicine and global health.

Chapter 2. Supply of medical radioisotopes in nuclear medicine: Challenges and perspectives

Mo-99 is a critical isotope used in over 40 million medical procedures annually, making it essential that reliable and secure production be assured.

2.1 Current situation and challenges of medical radioisotopes production

Ageing reactors: Implications for medical radioisotope production

The global production of Mo-99 relies on a few ageing research reactors that were not initially designed for continuous isotope production. Some of these reactors, such as the BR2 reactor in Belgium and the High Flux Reactor (HFR) in the Netherlands, have been used since the mid-20th century. These facilities face several issues, including unplanned outages, which increase the risk of supply disruptions. In October 2024, the HFR was shut down due to a structural issue requiring immediate repair. This unplanned shutdown, combined with scheduled maintenance at other reactors, exacerbated the shortage of both Mo-99 and Tc-99m. Table 2 provides an overview of the current research reactors used for medical radioisotope production, detailing the reactor name, country, year of commissioning, and their respective production capacities.

Research reactors like the above were not initially designed to produce isotopes such as Mo-99 on a large scale. As a result, their operational efficiency and reliability can be limited when adapted for this purpose. Additionally, older infrastructure often suffers from deterioration, increasing the likelihood of equipment malfunctions, unplanned outages, or the need for maintenance for extended periods. Furthermore, securing spare parts for older systems can be challenging, as many components are no longer be manufactured or readily available.

Table 2 Research reactors with significant Mo-99 production capacity

Reactor name	Location	Year commissioned	Approximate capacity (6-day Curie/week)
BR2	Mol, Belgium	1961	8 600
SAFARI-1	Pelindaba, South Africa	1965	6 200
LVR-15	Řež, Czechia	1957	3 200
MARIA	Świerk, Poland	1974	3 000
HFR	Petten, Netherlands	1961	3 000
OPAL	Lucas Heights, Australia	2006	2 200

Table 2 only presents sources of production where capacity is greater than 2000 6-day Curie per week. Approximately 95% of the global Mo-99 supply comes from the reactors listed in the Table above (Ferruci et al, 2022). Some other sources of production exist, for example from commercial reactors and even non-reactor technologies; however, these are presently limited. These additional and limited sources supply principally domestic markets, although they could be scaled up in the future.

Transitioning from highly enriched uranium (HEU) to low-enriched uranium (LEU)

A global initiative to reduce HEU usage and convert reactors from HEU to LEU has impacted medical radioisotope production and supply chains in recent years. This transition is driven by non-proliferation objectives, since low enriched uranium cannot be used for weapons.

The HEU to LEU transition applies at two levels during the process of isotope Mo-99 production: reactor fuel and irradiated targets. These two transitions do not necessarily occur at the same time; they can take place at different times, because they involve distinct technical and operational considerations.

Several reactors have already transitioned to LEU fuel use at the reactor fuel level. The LVR-15 reactor in Czechia switched to LEU fuel in 2011 (Koleska et al., 2011), while Poland's MARIA reactor switched in 2014 (Migdal and Krok, 2014). Australia's OPAL reactor, commissioned in 2006, was designed to use LEU fuel from the outset, making it a pioneer in this field. On the other hand, the BR-2 will continue to employ some HEU until a planned complete phase-out in 2026 (SCK CEN, 2023).

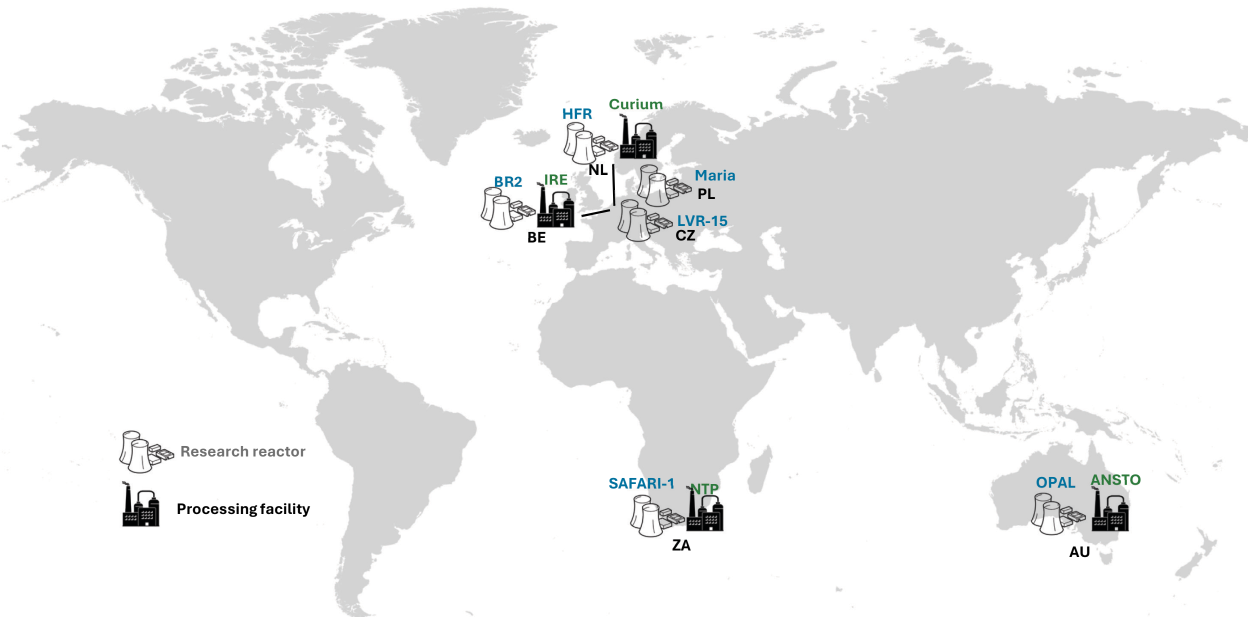
At the target level, the HEU to LEU transition has advanced significantly, with major producers like the Netherlands' Curium Pharma and South Africa's NECSA completing their conversion to LEU targets by 2017 and 2018, respectively. Poland's MARIA reactor followed suit in 2018 (Norwegian Ministry of Foreign Affairs, 2018) under an agreement with Curium Pharma. Finally, The Institute for Radioelements (IRE) in Belgium has successfully converted to LEU use since early 2023 (IRE, 2023).

In general, transitioning targets from HEU to LEU has been challenging for medical radioisotope producers at a technical level due to the lower efficiency of LEU targets, which yield less Mo-99 per irradiation cycle and result in more radioactive waste. Despite these associated operational costs, transitioning to LEU for reactor fuel and irradiated targets represents a critical milestone in enhancing nuclear security.

Global supply chain risks: International considerations

The global supply chain for Mo-99 and its decay product Tc-99m, remains critically vulnerable to geopolitical events and logistical challenges. These isotopes rely on a complex and highly interconnected supply chain characterised by just-in-time delivery (Figure 3). The COVID-19 pandemic exposed the fragility of this system as restrictions in air travel disrupted the transportation of isotopes, with greater negative impact on those with short half-lives. Similarly, geopolitical tensions, natural disasters, and trade restrictions can threaten the security of the supply and the on-time delivery of these isotopes.

Figure 3 Map of major research reactors and processing facilities for Mo-99 production



The concentration of production capacity in a few regions (only six research reactors) further exacerbates these risks. For example, the unavailability of European reactors in late 2022 due to planned and unplanned outages resulted in a near-complete halt in European processing capacity, illustrating the need for greater regional diversification. This dependence on European infrastructure underscores the necessity of collaborative efforts to mitigate these risks. These include integrating new production facilities into the global network and fostering international collaboration. For instance, the PALLAS reactor project in the Netherlands aims to stabilise the European supply chain, enhancing the security of supply for Mo-99/Tc-99m (NRG, 2024). Outside Europe, Japan has taken proactive steps to reduce its reliance on international supply chains by aiming to produce part of its Mo-99 needs domestically by 2028 (Naoyuki et al., 2023). Such projects exemplify efforts to diversify capacity and reduce dependence on a limited number of facilities.

The transport of medical radioisotopes poses additional challenges, particularly as regulations evolve. Changes to safety standards complicate logistics and necessitate continuous adaptation by producers and distributors. Addressing these transport vulnerabilities is crucial to maintaining the reliability of the supply chain.

Production costs

As previously noted, the production of Mo-99 has seen increased costs due to the global transition from HEU to LEU (Cutler and Schwarz, 2014). This transition is undertaken for non-proliferation purposes and requires significant investments in converting reactors and processing facilities. While many existing producers have absorbed these costs, they contribute to overall operational expenses in a market experiencing high pressure to lower prices. This dynamic is further complicated by the ageing infrastructure used for Mo-99 production. Many historical reactors and processors require upgrading or replacement to remain operational. Such infrastructure investments deter new producers since they must compete in a market where current producers sometimes receive what could be characterised as state subsidies that obscure the actual cost of production. These are factors that can disincentivise private sector participation.

Additionally, Mo-99 production is costly due to regulatory rules and transportation logistics. Mo-99 has a short half-life. Therefore, it must be produced and distributed in a very efficient manner. This adds to the complexity and requires continuous producer investment and demanding management to secure supply security. The recent decision by NorthStar Medical Radioisotopes to halt Mo-99 production exemplifies these challenges. NorthStar attributed its decision to withdraw due to rising production costs and competition from foreign government-subsidised producers. This illustrates how an inefficient market can lead to the exit of new suppliers, further exposing the fragility of the global supply chain.

2.2 Supply of lutetium-177 for radioligand therapy

The medical radioisotopes community has generally managed to mitigate vulnerabilities to the security of supply of diagnostic isotopes. However, with the emergence of RLT similar efforts will be required to guarantee a consistent and reliable supply of new therapeutic radioisotopes like lutetium-177 (Lu-177). Lu-177 has emerged as a key radioisotope in RLT because of its beta-emitting radiation. It has a half-life of 6.7 days, and its energy can be used to cause cancer cell damage while the limited range of beta particles ensures the action is confined to the tumour site and spares surrounding healthy tissue. Despite its importance, several issues in the Lu-177 supply chain must be addressed to meet increasing clinical demand and ensure equitable access.

One of the biggest challenges for the Lu-177 supply is the dependence on enriched ytterbium-176 (Yb-176), which is used as a target for Lu-177 production. Historically, Russia remains the primary source of enriched Yb-176, putting the supply of Yb-176 into the global market at risk due to uncertainties in the current global trading order.

Another key issue is scaling up the technically demanding production of non-carrier-added (NCA) Lu-177 compared to carrier-added Lu-177, which is easier to produce. Carrier-added Lu-177 is made from natural lutetium but is less pure and contains a contaminant, lutetium-177m, which is relatively inefficient for therapeutic applications. On the other hand, NCA Lu-177, produced from enriched Yb-176, is highly pure, with high specific activity and no impurities, making it efficient for treating small tumours since only a small quantity is needed. Producing NCA Lu-177 is a rather complicated process that involves chemical separation techniques and the reuse of enriched Yb-176, which raises the cost of production.

Another challenge is to reduce reliance on the research reactors for the same reliability reasons outlined earlier in the present report. Attempts to increase production at power reactors such as CANDU reactors in Canada have enhanced robustness. However, these systems are not yet online as they are still being optimised to meet global demand.

Finally, the post-irradiation separation of Lu-177 from Yb-176 is a complex and expensive process. The recycling of Yb-176 is very important because it helps reduce the cost of production by reducing the amount of waste generated, but this requires dedicated infrastructure and qualified staff.

Table 3 Summary of major novel medical radioisotope production projects and technologies

Country	Project name	Type	Primary sotopes	Operational timeline	Key features
Belgium	MYRRHA	Accelerator-driven system	Mo-99	2027	First liquid-metal reactor with proton bombardment
Canada	CANDU Reactor (Bruce Power)	Power reactor with isotope production	Lu-177	Lu-177 production doubled since 2022, with further expansion planned by 2027	Scalable irradiation system for Lu-177; integrated with power generation without outages
Canada	CANDU Reactor (Darlington Nuclear Generating Station)	Power reactor with isotope production	Mo-99, Tc-99m, Lu-177, Y-90	Production expected to start in 2025	Mo-99: 7-day cycle, 50 runs/year; Y-90: 3.5-day cycle, 50-52 runs/year; Lu-177: 7-14 day cycle.
Canada	TRIUMF Cyclotron	Accelerator	Tc-99m	Approved by Health Canada in November 2020	Direct Tc-99m production from Mo-100 via proton bombardment, eliminating the need for reactor-produced Mo-99.
France	Jules Horowitz Reactor (JHR)	Research reactor	Mo-99, Lu-177, Y-90	2032	25-35 day cycles, advanced cooling & unloading
Netherlands	PALLAS Reactor	Research reactor	Mo-99	Early 2030s	55 MW, LEU fuel, 300 days/year
Poland	MARIA Research Reactor	Research reactor	Mo-99, Lu-177	Beyond 2027	Extending operational life, cyclotron enhancement
United States	SHINE Technologies – Chrysalis	Accelerator	Mo-99	2027	5½-day cycles, 50 weeks/year; LEU solution irradiation via fusion accelerators; independent units for reliability and redundancy
United States	NorthStar Electron Accelerator	Reactor and accelerator	Mo-99, Cu-67, Ac-225	Mo-99 (2018 -2023)	6.5 days per week, 52 weeks per year using enriched >98% Mo-98 target material; minimal waste, mainly short-lived DIS products (Zr-95, Nb-95).
Argentina	RA-10 Reactor & Fission Radioisotope	Open-pool reactor & processing facility	Mo-99, I-131	2024(RA-10) + processing facility expansion	Multiple irradiation positions, LEU targets, Good Manufacturing Practice (GMP) facility

2.3 Resilience strategies for Mo-99

New production facilities: Transitioning to a modern and sustainable infrastructure

Several countries are investing in modern, sustainable infrastructure to address the challenges presented above, enhance production capacity and ensure supply resilience. The most significant projects are summarised in Table 3[2].

[2] The order of presentation reflects a combination of factors, including expected delivery timelines, potential impact, and the level of technological innovation involved. This approach is intended to illustrate the diversity and complementarity of these efforts, without implying any hierarchy or preference among projects.

The PALLAS reactor, currently under construction in Petten, the Netherlands, will replace the ageing HFR and help enhance the security of supply for medical radioisotopes. Designed as a 55-megawatt open pool reactor, PALLAS incorporates advanced safety features and uses LEU, aligning with international standards. Supported by EUR 2 billion in state aid, construction milestones, including foundation work, are well underway, with the reactor expected to become operational by the early 2030s. Once operational, PALLAS will run 300 days annually, supporting the treatment of approximately 30 000 patients daily worldwide (NRG, 2024).

France's Jules Horowitz Reactor (JHR) is expected to be operational by 2032 with the capacity to meet between 25% and 50% of Europe's annual demand for Mo-99 and to produce other isotopes for medical purposes, industry, research and development (JHR Material Test Reactor, 2025). JHR advertises the use of four displacement systems, including automated cooling and loading-unloading mechanisms, designed to ensure efficient production of radioisotopes (Mo-99 in priority). JHR's production cycle will operate in 25 to 35-day cycles, with up to six cycles per year. During these cycles, target materials are irradiated in dedicated locations within the reactor core and reflector to produce medical radioisotopes such as Mo-99, Lu-177 and yttrium-90 (Y-90).

Canada has harnessed the unique capabilities of its CANDU reactor technology, which is fuelled with natural uranium, to serve a dual purpose, integrating medical radioisotope production into its power-generating infrastructure. With their distinctive feature of online refuelling, CANDU reactors enable continuous isotope production without disrupting electricity generation. Currently, cobalt-60 is extracted from reactors at Pickering Nuclear Generating Station and Bruce Power's Bruce B plant every 24 to 30 months (OPG, 2025a). The refurbished Darlington Unit 1 was producing cobalt-60 as of January 2025, and Ontario Power Generation (OPG) announced plans to modify Darlington's three other units to allow them to also produce the isotope (OPG, 2025b). Ontario's CANDU reactors produce around 50% of the world's supply of cobalt-60.

CANDU's unique design with high neutron flux, online refuelling capabilities, and low pressure and temperature allow for extended run-times between shutdowns, enabling the production of other isotopes like Mo-99 and Lu-177, and Y-90. The Darlington Nuclear Generating Station and Bruce Power are excellent examples of this innovation. At Darlington, the Target Delivery System (TDS), an advanced in situ system developed by Laurentis Energy Partners and BWXT Medical has been installed and tested for Mo-99 production and is currently pending approval from the US FDA and Health Canada (Laurentis, 2023). The arrangement between Laurentis and BWXT Medical is expected to produce enough Mo-99 to supply a "significant portion" of current and future North American demand of technetium-99m, the decay product of Mo-99 used in more than 40 million medical treatments each year (BWXT, 2023; Laurentis, 2023). At Darlington, Mo-99 production operates on a 7-day cycle, with 50 planned runs per year, while Y-90 follows a 3.5-day cycle and runs 50-52 weeks annually. Lu-177 production cycles range from 7 to 14 days. Unlike conventional uranium-based Mo-99 production, Darlington employs natural molybdenum washers as the target material, minimising waste and proliferation concerns. Y-90 production utilises Yttrium Oxide (Yb_2O_3), and Lu-177 production is based on ytterbium oxide (Yb_2O_3) irradiation. Efficient isotope handling is essential due to the rapid decay of medical radioisotopes. To address this, Laurentis's TDS is designed for seamless, automated isotope transfer, minimising operational delays. Equipped with four target positions and active cooling, the system provides unmatched irradiation capacity and flexibility for producing a range of medical radioisotopes. Currently, the TDS is irradiating Mo-99 for BWXT Medical and is expected to begin Y-90 production for Boston Scientific in late 2025, pending regulatory approvals. Additionally, Lu-177 production is planned to start in 2025, further expanding Darlington's contribution to critical cancer treatments.

Bruce Power, meanwhile, as part of an international collaboration with the Saugeen Ojibway Nation (SON), Isogen (a Kinectrics and Framatome company), and ITM Isotope Technologies Munich SE (ITM) has implemented a novel Isotope Production System (IPS)

(Bruce Power, 2022), doubling Lu-177 production capacity since 2022, with further expansions planned by 2027 (Bruce Power, 2024).

In Belgium, SCK CEN is developing MYRRHA (Multi-purpose hYbrid Research Reactor for High-tech Applications), an experimental irradiation facility designed to replace the BR-2 reactor. MYRRHA will be the world's first liquid-metal-cooled nuclear reactor powered by a particle accelerator. The facility is expected to begin Mo-99 production in 2027 and will introduce a new generation of medical radioisotopes by using proton bombardment instead of neutron activation. Planned Mo-99 production cycles in MYRRHA will last approximately 168 hours (7 days), slightly longer than BR-2's 150-hour (6.25 days) cycle (Buglione, 2016; MYRRHA, 2024).

In the United States, SHINE Technologies is pioneering an accelerator-driven fusion-driven approach to isotope production at its new facility, the Chrysalis. This innovative method eliminates the need for traditional nuclear reactors, providing an alternative for Mo-99 production. Expected to begin operations by 2027, the facility is designed to supply more than one-third of the global demand (SHINE Technologies, 2024).

As an alternative and innovative approach to traditional fission-based Mo-99 production, TRIUMF in Canada is developing a cyclotron-based method that directly produces technetium-99m (Tc-99m) from Mo-100 via proton bombardment. This method eliminates the need for reactor-produced Mo-99, which would allow hospitals with advanced cyclotrons to produce Tc-99m locally, reduce supply chain vulnerabilities, and eliminate radioactive waste concerns. The approach was approved by the Government of Canada Ministry of Health for human use in November 2020. TRIUMF's system enables medical cyclotron centres to achieve an end of synthesis daily production capacity of approximately up to 22 000 mCi/batch of Tc-99m, enabling on-demand, regional production and further reducing dependence on centralised Mo-99 supply chains (ARTMS, 2020). This technology was licensed to ARTMS, a spinoff company acquired by Telix Pharmaceuticals in April 2024.

Similarly, NorthStar Medical Radioisotopes introduced an innovative non-uranium reactor approach to Mo-99 production using Mo-98 target material, which was approved by USFDA in 2018. A second production approach using electron accelerator technology, using Mo-100 as a target, is in development (NORTHSTAR, 2023). These methods eliminate the need for uranium and generate significantly less radioactive waste. However, NorthStar halted Mo-99 production in 2023, citing economic viability concerns linked to competition from government-subsidised foreign producers, reimbursement challenges in the United States and rising operational costs. Since then, NorthStar has shifted its focus to the production of novel radioisotopes for RLT, including routine production of copper-67 (Cu-67) for clinical trials and plans to initiate NCA actinium-225 (Ac-225) production at a commercial scale (NORTHSTAR, 2024).

When Mo-99 production was active (2018–2023), NorthStar operated on a 6.5-day production cycle per week, running 52 weeks per year. The process utilised highly enriched (>98%) molybdenum-98 target material, ensuring efficient isotope yield while minimising waste. The primary radioactive byproducts consisted of short-lived decay-in-storage products, such as small quantities of zirconium-95 and niobium-95, allowing for a more manageable waste profile compared to traditional uranium-based production methods.

Poland is extending the operational life of the MARIA Research Reactor beyond 2027 and installing new facilities, such as a cutting-edge cyclotron, to enhance its production capabilities further(NCBJ, 2024). Typical production cycle is 6-7 days (rarely up to 9 days), number of cycles per year varies between 30 and 40, depending on the scheduled maintenance breaks. On average, MARIA operates around 4 800 hours per year. Its design allows the installation of various types of irradiation facilities and neutron sources corresponding with the neutron flux performance 2×10^{14} neutrons/cm^{2}s of thermal neutrons and 3×10^{13} neutrons/cm^{2}s of fast neutrons. In the reactor core and reflector area there are a number of vertical irradiation channels, mainly used for radionuclide production. Various target materials are irradiated. Radionuclides typically produced in the MARIA research reactor are I-131, Lu-177, samarium-153, holmium-166 and others. In terms of the radioactivity volume, the I-131 produced either via neutron irradiation of natural tellurium (Te) or tellurium enriched in Te-130 as the target material is on top of the list. Irradiation of uranium targets for Mo-99 production has been carried out in the MARIA reactor since 2010. Adaptation of the reactor fuel channel to irradiate uranium plates relies on its modification to provide multiple loading and discharge of uranium targets into and out of the installation without the need to evacuate the whole irradiation channel from the position socket in the reactor core. MARIA contributes to the global supply chain of Mo-99 with its irradiation at the level of 1.23 PBq/year.

Finally, Argentina is developing the RA-10 reactor to meet market demand in South America. Designed as a 30 MW open pool reactor using LEU fuel, RA-10's production cycle includes many positions for irradiating uranium/aluminium LEU targets for Mo-99 and I-131 production. The RA-10 will operate in a production cycle of 29.5 days – 28 days for irradiation and a stop of 1.5 days, operating 46 weeks per year, with 6 weeks reserved for maintenance.

Since 1985, Argentina has the Fission Radioisotope Production Plant, a facility for processing up to 250 Ci of Mo-99 (6-day end of processing). This facility runs one batch per week with the targets irradiated in the RA-3 reactor, operating 46 weeks per year with 6 weeks reserved for maintenance. Argentina is also developing a new Fission Radioisotope Production Plant, a GMP facility for processing targets irradiated in RA-10 reactor with a weekly production capacity of 1 250 Ci of Mo-99 (6-day end of processing) in several batch processes (CNEA, 2024).

Public investment and market competitiveness

Public investment is often essential for supporting certain high-impact initiatives that the private sector cannot undertake alone, such as the deployment of nuclear reactors or research and development (R&D) in innovative technologies. However, persistent subsidisation distorts markets, discourages competition, and crowds out private investments, resulting in negative economic benefits. Consistent with this rationale, HLG-MR policy recommendations have included the implementation of the full-cost recovery (FCR) principle, ensuring that the pricing of Mo-99 reflects the actual cost of production, including capital investment, operational expenses and waste management. This approach aims to create a sustainable and economically viable supply chain, reducing reliance on government support while fostering market-driven investment in new medical radioisotope production facilities. The success of these new projects in building supply chain resilience and diversification will depend on both economic viability and ensuring they can reliably meet the growing demand for medical radioisotopes.

One of the challenges of building a robust and economically sustainable supply chain is that many medical radioisotopes cannot be stored for long periods of time. In addition, their marginal cost of production is far lower than their average cost of production, the latter including an allowance for fixed capital costs. This means that one mechanism through which producers can break even is by restricting capacity to a point where periodic supply interruptions generate price peaks during which they recuperate their fixed capital costs; however, other mechanisms may also contribute to covering these costs. In other words, in the current organisation of the market, even theoretical FCR implies high price volatility and periodic price interruptions. In practice, such price volatility induces producers to restrict capacity even further. In such a situation, which is experienced in other markets for non-storable goods (electricity, bandwidth, among others), one way towards a more sustainable supply and smoother pricing is to remunerate producers for capacity, which will occasionally be left unused, as well as for production, i.e. the finished radioisotope dose. Such an arrangement would not be without its challenges - in particular because the capital costs of different producers differ widely, say from building a dedicated new plant to adding some extra equipment to an already running nuclear power plant, and information asymmetries are high. Yet, one could start with fixed capacity payments for all, whose cost would be included in the price of every single dose and see how the market develops. Such capacity payments could exist with otherwise forthcoming support at the national level.

The Canadian Medical Isotope Ecosystem (CMIE) serves as a practical example. In 2023 the Canadian government allocated up to CAD 35 million over five years to the CMIE to develop initiatives focused on the production, advancement and distribution of medical radioisotopes and radiopharmaceuticals in Canada. This funding will allow CMIE to support projects with Bruce Power, BWXT Medical, McMaster University and Canadian Nuclear Laboratories. The CMIE will operate under the oversight of the Centre for Probe Development and Commercialization (CPDC) and, TRIUMF Innovations(Government of Canada, 2023). As mentioned, nuclear utilities such as Bruce Power effectively combine electricity generation with medical radioisotope production, optimising reactor use. This dual-purpose model allows facilities to contribute to growing demand while generating additional revenue that helps offset the associated operational expenses.

The PALLAS project provides a recent example of how public funding can support the development of novel infrastructure while adhering to FCR principles, whereby pricing reflects production costs. This aligns with EU guidelines for ensuring fairness in market-based operations and EU financial transparency and competition principles. The European Commission's role involved evaluating the public funding model for PALLAS to ensure compliance with state aid rules while supporting the public policy goal of infrastructure development in nuclear medicine (EC, 2025). While PALLAS is designed to align with market-based principles by incorporating FCR into its pricing model, ensuring long-term financial sustainability remains a key consideration as the project progresses.

Reimbursement schemes play a crucial role in the FCR framework by influencing cost allocation, supporting investment recovery, and encouraging innovation. These schemes help structure financial flows between stakeholders, contributing to more predictable market operations. The interaction between public investment and market principles shapes industry dynamics. FCR intends to create a framework supporting long-term stability in the isotope market by structuring cost recovery mechanisms that account for private and public investments.

While the FCR principle aims to create an efficient market for medical radioisotopes, its implementation remains complex and debated. Some stakeholders advocate for a swift transition to FCR to reduce reliance on subsidies and encourage private investment. In contrast, others caution that abrupt changes could increase costs and limit patient access. Achieving FCR requires a balanced approach that fosters financial sustainability without compromising affordability and public health policy priorities in isotope production. Moreover, even among experts and policymakers who endorse the FCR principle, there are debates and disagreements about how to implement FCR. Further dialogue and policy refinement will be essential to addressing the challenges and opportunities associated with FCR.

Demand update and outage reserve capacity (ORC)

The global demand for Mo-99 continues to grow steadily. Emerging markets are experiencing a demand growth rate of 5% annually, while mature markets have a modest growth of 0.5% per year (NEA, 2023). This growth reflects expanding health care access in developing regions and the increasing use of nuclear medicine for diagnostic and therapeutic purposes. However, these trends also underscore the importance of addressing structural challenges in the Mo-99 supply chain, mainly as it operates on a just-in-time basis with limited capacity for stockpiling due to the isotope's short half-life.

As noted, unplanned outages at major reactors, especially in Europe, can quickly lead to global shortages. Ensuring this extra capacity – termed outage reserve capacity (ORC) is crucial for meeting fluctuations in demand and unexpected outages. However, achieving and sustaining this level remains a persistent challenge due to the challenges presented in Section 2.1.

2.4 Resilience strategies for Lu-177

North American companies are actively working to develop domestic sources of ytterbium-176 (Yb-176) to reduce reliance on Russian supplies for Lu-177 production. Unlike uranium fuel enrichment, which involves increasing the concentration of fissile isotopes like U-235, the enrichment of Yb-176 refers to the process of improving the proportion of Yb-176 within naturally occurring ytterbium, which contains a mix of stable isotopes. In this context, SHINE Technologies and Kinectrics have made significant progress in advancing isotopic enrichment technologies tailored for Yb-176 (SHINE Technologies, 2022; KINECTRICS, 2024).

Investing in novel enrichment technologies and initiatives, such as those by the CMIE, has improved the supply chain and offers scalable solutions. A notable example is the partnership between Bruce Power and Isogen, which has led to the installation of a novel IPS in Unit 7 of the Bruce Nuclear Generating Station. This installation marks the first time a power reactor has been equipped to produce Lu-177, enhancing the reliability of its supply. Additionally, new approaches like SHINE's neutron source facilities in the United States are paving the way for alternative production methods that do not rely on traditional nuclear reactors, thereby strengthening the supply chain. Their method utilises a low-energy, accelerator-based neutron source to irradiate enriched ytterbium-176 (Yb-176) targets (World Nuclear News, 2019).

Another critical area is strengthening the separation and recycling processes for the enriched Yb-176. Efficient use of resources and low costs can only be achieved through advanced chemical separation technologies and recycling programmes (Horwitz et al., 2005). Public and private investment in R&D on these processes could benefit the economics for Lu-177 production.

These strategies can help meet the growing demand for radioligand therapies.

2.5 Waste management, transportation and radiological protection

While waste management, transportation, and safety regulations for medical radioisotopes have been extensively studied and refined over decades – mainly due to the widespread use of Mo-99 – the emergence of RLT introduces new challenges. RLT relies on radioisotopes with different half-lives and radiation characteristics (see Table 1), necessitating the adaptation and expansion of existing regulatory frameworks. These isotopes are relatively new in medical applications. Regulatory bodies can help markets and health care systems operate efficiently by developing and harmonising guidelines to address waste disposal requirements, logistical constraints related to transportation and the half-life of medical radioisotopes, and radiological protection measures.

The production of medical radioisotopes generates radioactive waste, which must be appropriately managed to make sure that radiological risks to people and the environment are kept as low as reasonably achievable (ALARA). The transition from HEU to LEU targets in Mo-99 production presents challenges in waste management, as LEU targets generate larger volumes of radioactive waste. Efforts are being made across various production facilities to address these challenges and optimise waste-handling strategies. Due to regulatory requirements, the industry must advance processing technologies to enhance efficiency while minimising waste generation.

Waste management also applies at the end of the lifecycle of use. Waste from RLT consists primarily of patient excreta that enters wastewater systems, necessitating monitoring for long-term radiological impacts. Studies on Lu-177 discharge indicate that radiation levels remain within allowable limits; however, with the increased use of RLT, continued assessments will be necessary to maintain environmental safety. In addition to patient waste, hospitals using RLT produce contaminated medical supplies, such as syringes, gloves, and protective garments, which require strict decontamination and disposal procedures to minimise radiation exposure of health care workers and prevent contamination of medical facilities.

The transportation of medical radioisotopes is complex due to their short half-lives and the strict international shipping regulations governing their movement. The supply chain involves multiple stakeholders – including research reactors, processing facilities, generator manufacturers, and transport companies – making it highly susceptible to disruptions. For the case of Mo-99 transporting irradiated uranium targets requires Type B(U) containers, which must be specially approved for safe transport in compliance with established safety standards. On the other hand, air transport constraints, regulatory differences among countries, and lengthy customs procedures continue to pose significant challenges in maintaining a stable isotope supply chain. The European Observatory on Medical Radioisotopes continues to serve as a key actor in co-ordinating logistics and implementing rapid response strategies to prevent supply chain bottlenecks.

Given the short half-life of Lu-177 (6.7 days), efficiency in logistics is critical since any delay in the delivery of the product can lead to a reduction in its effectiveness. The co-ordination between suppliers and treatment centres requires careful scheduling to maximise the utilisation of each batch before significant radioactive decay occurs. It should also be taken into consideration that Lu-177 is a beta emitter with low-energy gamma emissions, requiring moderate shielding which has positive impact on transportation cost and flexibility compared to the situation outlined above for Mo-99 which is a gamma emitter.

Much of the existing infrastructure was not designed for the production, transport and application of therapeutic isotopes. As markets for new medical radioisotopes emerge, infrastructure requirements and regulatory requirements associated with transport and containment may change, in some cases becoming more stringent. For example, alpha-emitting radionuclides will likely require more stringent transport conditions due to their higher radiation levels. Producers will benefit from closely monitoring regulatory frameworks as well as the IAEA's authorisation standards for containers.

All medical radioisotope supply chain stakeholders must comply with strict safety regulations during production, handling and transport. The European Commission – Directorate-General for Energy, Radiation Protection and Nuclear Safety, ensures oversight of radiological protection measures and monitors the national adoption of basic safety standards in accordance with Council Directive 2013/59/Euratom of 5 December 2013, which lays down basic safety standards for protection against the dangers arising from exposure to ionising radiation. Safety protocols cover worker protection, facility security, and radioactive waste management, aligning with IAEA, US Nuclear Regulatory Commission (NRC), and Euratom guidelines. As the demand for medical radioisotopes continues to grow, enhanced international co-operation is needed to streamline regulations, monitor compliance, and ensure the uninterrupted availability of these essential products in the health care sector across borders.

Ensuring a harmonised and well-regulated radiological protection framework is essential as RLT use grows. Regulatory bodies, health care institutions, and nuclear medicine stakeholders will help create a more efficient market by working together to establish standardised or harmonised international guidelines for outpatient RLT, radiopharmaceutical waste disposal protocols, and personalised dosimetry approaches to optimise patient care while minimising unnecessary radiation exposure. Given the increasing use of Lu-177-based therapies, safety measures in clinical settings will also require radiological protection for health care professionals administering RLT. Proper training in handling radiopharmaceuticals, dedicated treatment rooms, lead shielding, and personal protective equipment (PPE) must account for occupational exposure. A proactive and co-ordinated approach will ensure that radiological protection measures keep pace with medical advancements while maintaining high safety standards for patients and health care workers.

Chapter 3. Radioligand therapy for cancer treatment

RLT has been identified as a promising treatment modality within the evolving field of precision medicine. Combining the power of nuclear technology with molecular specificity, RLT provides a highly selective treatment to locate and destroy cancer cells while minimising side effects on healthy tissue. While the use of radioisotopes in therapy dates back to the mid-20th century, modern RLT represents a significant evolution, offering active molecular targeting rather than relying on passive uptake, such as thyroid's natural affinity for iodine (as described in Section 1.2). This innovative approach is not only changing the way some types of cancer are treated but is also helping to address some unmet medical needs and giving new options to patients currently facing limited treatment options.

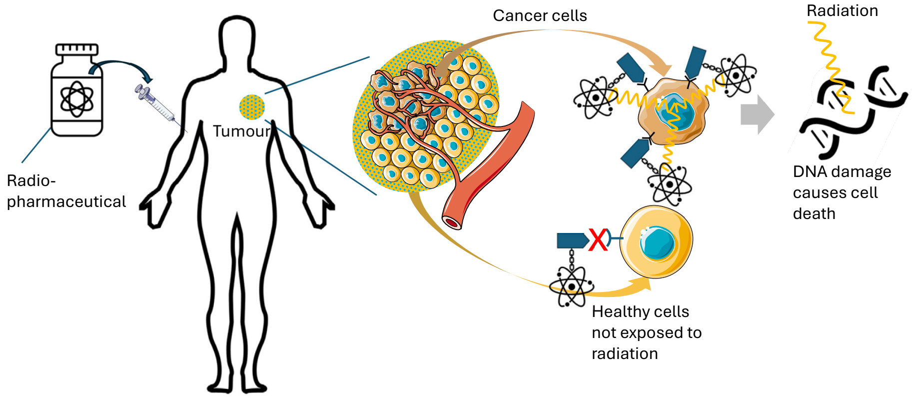
3.1 Significance of RLT for cancer treatment: An innovative therapeutic modality

RLT is built on decades of innovation in nuclear medicine, tracing its origins to the pioneering work with iodine-131 (I-131) in the mid-20th century. I-131, a radioisotope with therapeutic properties, was initially utilised for treating hyperthyroidism and, subsequently, thyroid cancer (Fahey and Grant, 2021). This marked the first successful application of radionuclide therapy in clinical practice, setting the foundation for integrating radioactive isotopes into cancer care. Over the years, advancements in nuclear medicine have led to more sophisticated therapies, transitioning from generalised approaches to precision-targeted treatments, such as RLT.

RLT leverages molecular specificity to achieve specificity in cancer treatment. It relies on radioligands – molecular complexes composed of a radioisotope paired with a ligand. A ligand is a biological molecule designed to bind selectively to specific features of cancer cells, such as surface proteins or receptors. This targeted mechanism allows the radioligand to deliver the radioactive payload directly to the cancer site, damaging the DNA. Figure 4 illustrates the step-by-step process of RLT, from drug preparation to targeted cancer cell destruction.

The therapeutic action of RLT stems from the radiation emitted by the radioisotope, which damages and destroys cancer cells. Different radioisotopes are chosen based on their decay times and spatial range of action, enabling tailored treatment approaches for various cancer types. This specificity is a critical advancement in oncology, as it focuses therapeutic action exclusively on malignant cells, enhancing both efficacy and safety (Sgouros et al., 2020).

Figure 4 Step-by-step process of RLT, from drug preparation to targeted cancer cell destruction



Note: From left to right: a vial containing the radioligand; administration of the drug via injection into a patient with a tumour; localisation of the radioligand within the tumour and its surrounding vasculature; selective binding of the radioligand to cancer cells expressing the target receptor while sparing healthy cells; and finally, DNA damage induced by radiation, leading to cancer cell death.

The development and adoption of therapies like Lu-177 PSMA for metastatic castration-resistant prostate cancer (mCRPC) highlights the transformative potential of RLT. Prostate cancer, a significantly more common condition than neuroendocrine tumours, whose treatment has also benefited from RLT, has become a primary focus of RLT research and application.

Impact on health care systems and patient benefits

RLT's ability to precisely target cancer cells addresses critical challenges in oncology, particularly for cancers resistant to conventional therapies. Its specificity reduces the burden of collateral damage, improving the quality of life for patients undergoing treatment. Furthermore, the success of RLT in managing prostate cancer has spurred interest in expanding its applications to other cancers, including multiple myeloma, malignant melanoma, pancreatic cancer, breast cancer and lung cancer (Uijen et al., 2021).

As radiopharmaceuticals represent a step change in cancer treatment, their integration into clinical practice underscores the need for health care systems to adapt and expand infrastructure to support this innovative therapy (Czernin and Calais, 2022; Herrmann et al., 2022). Should patient volumes and doses administered grow, there will be a need for increased production capacity and equitable access to ensure that RLT reaches all eligible patients globally.

3.2 Approved drugs and clinical trials: current progress and prospects

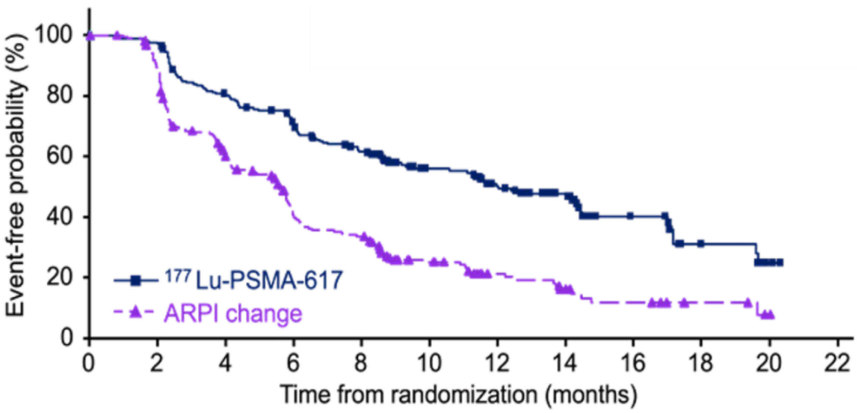
Recent developments in the field of RLT have included the approval and commercialisation of Lu-177 DOTATATE and Lu-177 vipivotide tetraxetan, two radiopharmaceuticals that have transformed cancer treatment paradigms. These therapies rely on Lu-177, which emits beta radiation with lethal effects on tumour cells while sparing surrounding healthy tissue. Their success has established the viability of RLT as a highly selective and effective treatment modality, paving the way for more targeted therapies (Zhang et al., 2020).

Lu-177 DOTATATE, approved in 2018, targets gastroenteropancreatic neuroendocrine tumours. As the first RLT approved for cancer treatment, it marked a significant milestone in the field. Lu-177 vipivotide tetraxetan, approved in 2022, focuses on mCRPC. Given the high prevalence of prostate cancer globally (World Cancer Research Fund, 2022), Lu-177 vipivotide tetraxetan has rapidly gained traction. Industry forecasts indicate that its treatment volumes could surpass other RLT drugs (LLP, 2024).

Building on the success of Lu-77 DOTATATE and Lu-177 vipivotide tetraxetan, several ongoing clinical trials aim to evaluate additional applications and optimise their therapeutic potential (Uijen et al., 2021; Malandrino et al., 2024). These studies examine various aspects of RLT using Lu-177 and other novel medical radioisotopes for therapeutic applications, including dosage, combination therapies and use in early-stage cancer treatment.

Figure 5 presents the results from one such clinical trial (Oliver Sartor, 2023). In this trial, half of the patients received an androgen receptor pathway inhibitor (ARPI change) therapy, which is the standard therapy for treating prostate cancer. The other half received a novel RLT based on Lu-177 and the PSMA-617 ligand (Lu-177-PSMA-617). The vertical axis represents the percentage of patients who remained free from disease progression or death, while the horizontal axis shows the time elapsed following treatment. Results from this trial are promising, with those patients receiving RLT living longer without cancer worsening following treatment.

Figure 5 Clinical trial results for RLT based on Lu-177



Source: Oliver Sartor, 2023.

However, in the policy-making process, the assessment of RLT is not solely based on clinical outcomes. NICE in the United Kingdom, among other organisations, assesses new therapies not only for their efficacy in delaying disease progression and improving survival, but also for their impact on quality of life, side effects and the overall economic value for the health care system (National Institute for Health and Clinical Excellence, 2009). Cost effectiveness analyses, including treatment affordability, long-term patient benefits and health care system sustainability are key elements in determining RLT's accessibility and potential for widespread adoption.

Moreover, extensive efforts are being made to develop new RLT drugs targeting other cancers. These include basic and preclinical research and advanced clinical trials focused on identifying novel ligands for molecular targets beyond those addressed by existing therapies (Uijen et al., 2021; Malandrino et al., 2024).

While Lu-177 remains the cornerstone of RLT development, other research now explores the potential of alpha-emitting radioisotopes like actinium-225 and astatine-211 (Zuo et al., 2024). These isotopes offer shorter radiation ranges and higher energy emissions, potentially enabling even greater precision and efficacy in targeting tumours. Alpha particle-based therapies are particularly promising for tackling micrometastases and residual disease, areas where beta emitters may be less effective.

3.3 Theranostics: Benefits from the integration of diagnostics and therapeutics

Theranostics involves using radiopharmaceutical pairs known as theranostic pairs – where one compound is designed for imaging to identify the presence and abundance of a specific molecular target and the other for delivering targeted radiation therapy. To optimise the therapeutic outcome of RLT, accurate diagnostics are essential. Advanced imaging systems have made it possible to assess a patient's eligibility for treatment non-invasively and to maximise its impact. Imaging provides a holistic view of body physiology, and with molecular markers, it allows the extraction of critical information at the cellular and molecular levels.

Theranostic pairs enable clinicians to tailor treatment to individual patients by first confirming the presence of the therapeutic target. Occasionally, the same radiopharmaceutical can both identify the extent of the disease and deliver therapy in theranostics. However, separate but complementary compounds are most commonly used to achieve these dual objectives-these are the theranostic pairs.

An example of such a theranostic pair is presented below.

Step 1: Diagnostic imaging: A diagnostic tracer images the targeted peptide or protein. For instance, Gallium-68 (Ga-68) PSMA PET imaging measures the concentration of PSMA in prostate cancer patients. If the imaging detects a high expression, the patient will likely respond to targeted therapy.

Step 2: Therapeutic delivery: A corresponding therapeutic radiopharmaceutical, such as Lu-177 PSMA, is administered to deliver targeted radiation, destroying cancer cells while sparing healthy tissue.

This two-step process ensures that treatment is precisely directed to patients most likely to benefit, reducing unnecessary exposure and optimising therapeutic outcomes (Jeelani et al., 2014).

A notable development in the field is the recent clinical trial results exploring the use of theranostic agents as first-line therapies (Singh et al., 2024). This is the first time a theranostic approach has been proposed as the initial treatment option for certain cancers. This underscores the growing confidence in RLT's potential to transform cancer care by offering highly effective, personalised treatment options.

3.4 Description of a radioligand therapy session: Clinical workflow and patient experience

While this publication aims to support discussions and collaborations that strengthen the security of supply, it is helpful to understand the applications side and even take a glimpse into the doctor and patient experience. This section presents the clinical workflow and patient experience with RLT.

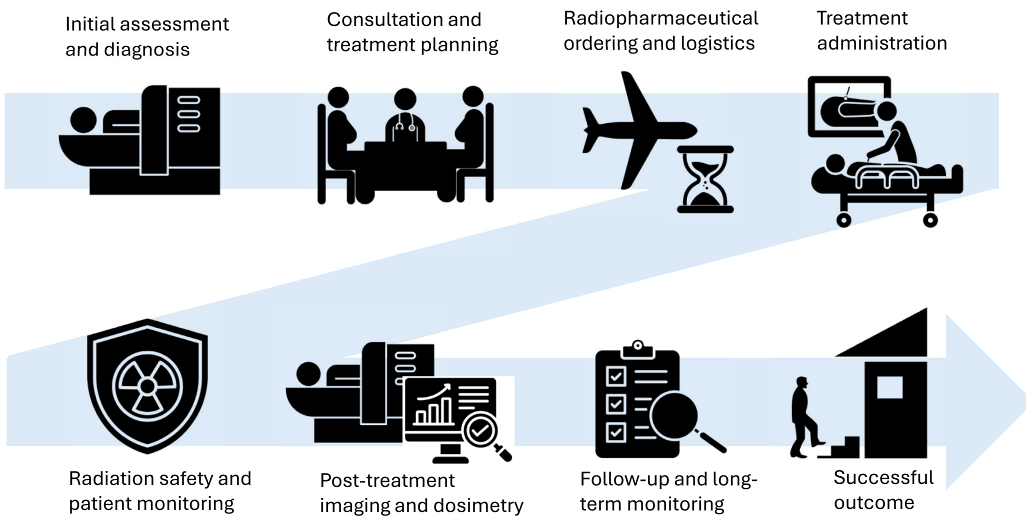
RLT begins with a multidisciplinary, patient-centred approach. An example of the workflow is shown in Figure 6. After a cancer diagnosis, the patient typically undergoes a PET/CT imaging exam to assess the tumour's molecular characteristics and determine eligibility for therapy. A nuclear radiologist interprets the imaging results and, alongside the oncology team, reviews the patient's medical history, lab results and prior imaging. If the findings suggest the patient is a good candidate for RLT, the team communicates with the referring oncologist to discuss the potential benefits and confirm the suitability of the therapy (George et al., 2024).

Once eligibility is confirmed, the patient meets with the care team, including nuclear medicine specialists, to receive a detailed explanation of the therapy process, its goals, and its potential risks and benefits. This personalised discussion ensures that the patient is fully informed.

Then, the treatment session is scheduled, typically requiring about two weeks to obtain the radiopharmaceutical from the manufacturer, given the need for precise customisation and logistical co-ordination. RLT therapy involves multiple sessions, usually 4-6, with several weeks of interval between sessions.

On the day of the therapy, the patient arrives at a specialised nuclear medicine facility or hospital equipped with a dedicated radiotherapy unit. The care team often includes nuclear radiologists, nuclear medicine technologists, and a radiation safety officer. The radioligand is administered intravenously or, in some cases, through direct injection into the targeted organ. The infusion process generally lasts 30 to 60 minutes, during which the patient is carefully monitored for immediate side effects (Halfdanarson et al., 2023).

Figure 6 RLT workflow



Rigorous radiological protection protocols are followed to ensure safety throughout the session. These include using shielded treatment rooms, PPE for staff, and strict waste disposal measures. After the infusion, the patient is hydrated to facilitate the elimination of radioactive material. Vital signs are continuously monitored, and the radiation safety team measures the patient's radioactivity levels. Before discharge, the patient must void their bladder, and their radiation levels must fall below a safe threshold determined by the radiation safety officer.

For therapies involving Lu-177-based drugs, an additional critical step consists of performing SPECT/CT scans following the treatment. These scans leverage the gamma rays emitted by Lu-177 to precisely assess the distribution of the radioligand within the patient's body. This process, known as dosimetry assessment, is crucial for planning subsequent treatment sessions. It helps to optimise the dose delivered to the tumour while minimising exposure to healthy tissues, ensuring the therapy's safety and efficacy.

After the session, the patient remains in a recovery area for observation. The duration of this phase varies depending on the radioligand used, the patient's condition and each country's safety guidelines and regulations. Once deemed stable, the patient is discharged with clear post-therapy instructions to minimise radiation exposure to others. Patient discharge protocols vary across countries, reflecting differences in safety regulations and health care protocols. In the United States, patients receiving Lu-177-based therapies are usually treated as outpatients and are allowed to go home the same day with thorough safety information (MSKCC, 2022). In Germany, where radiation safety rules differ, the patients may have to stay overnight in a special ward for as much as 48 hours to monitor radiation decay (UKB, 2025). This naturally impacts the number of patients treated weekly in each country. Thus, outpatient-based centres in the United States may have a higher patient throughput than facilities in Germany, affecting the economics of treatment facilities.

Patients are also provided with a follow-up imaging and consultation schedule to assess the therapy's effectiveness and overall impact on their health. In addition to PET/CT imaging, repeated SPECT/CT scans may be scheduled during the treatment course for ongoing dosimetry evaluations. These follow-ups are essential to monitor the tumour's response and make necessary adjustments to the treatment plan (Kazemi-Jahromi et al., 2024; Song et al., 2024).

Access to RLT, such as Pluvicto, varies across countries. While some countries – like Germany, Luxembourg, Slovenia and Brazil – have granted full regulatory approval and integrated the therapy into their national reimbursement systems, others have approved the drug but rely on more limited mechanisms. For example, France and Italy offer early access through government-backed programmes or patient-specific approvals, while countries like Austria and Switzerland provide partial coverage through diagnosis-related group (DRG) funding (Leonhard Schaetz, personal communication, 2025). Others have approved the therapies but have yet to implement them within reimbursement systems, delaying patient access. Meanwhile, in the majority of countries, these treatments remain unapproved, or the health care infrastructure is insufficient to integrate such an advanced technique. This technology is relatively new and wider adoption is expected in the coming years.

3.5 Infrastructure needs for enhancing diagnostic and therapeutic capabilities

The widespread adoption of RLT depends on various factors, including the infrastructure readiness of hospitals and private medical clinics (or speciality clinics), primarily within nuclear medicine departments. Since RLT involves using radioactive materials, this adds complexity and significantly increases the costs of establishing a dedicated therapeutic unit. Newer clinics often struggle with building the infrastructure for this workflow, particularly smaller practices and community hospitals, which may lack dedicated nursing teams or practice managers to co-ordinate patient care or adapt workflows-they lack the experience or infrastructure traditionally found in larger academic centres (Mittra et al., 2024).

The successful establishment of theranostics centres requires compliance with stringent regulatory, logistical and technical considerations. These include obtaining radioactive material licences, ensuring proper radiation shielding, and implementing protocols for radioactive waste management. Shielding requirements may range from polymethyl methacrylate (PMMA) storage boxes for vials to concrete bunkers for waste, dictated by the radionuclides' characteristics and national regulations (Herrmann et al., 2022).

Additionally, the operation of theranostic centres involves multidisciplinary collaboration among nuclear medicine specialists, medical physicists, radiopharmacists, and radiation safety officers. The complexity of theranostic procedures often necessitates specialised training and higher levels of expertise compared to diagnostic nuclear medicine. Facilities must accommodate increased activity levels for therapies and address logistical challenges like patient safety and waste management (Oncology Nurse Advisor, 2024).

The infrastructure also needs to support robust treatment planning and verification processes. Dosimetric evaluations, including whole-body and organ-specific assessments, are critical to optimising treatment efficacy and minimising radiation exposure. Integrating automated dispensing systems and imaging technologies like PET/CT is crucial in ensuring the precise delivery of radiopharmaceuticals.

Finally, with the growing global demand for theranostics, ensuring equitable access to these advanced treatments requires scaling up the availability of imaging and therapy facilities. Governments, professional societies, and international organisations could benefit from collaboration to provide education and training programmes, thereby addressing workforce shortages and enabling the seamless implementation of theranostic capabilities across diverse health care settings.

The RLT health care services can be divided into three categories: “experienced,” “initiated,” and “emerging,” based on their readiness. The first category comprises services or centres with advanced capabilities, established infrastructure, and significant expertise in RLT, enabling them to address diverse therapeutic needs. They are well-equipped with imaging systems, radio-protected chambers, and skilled personnel – often associated with research or clinical trial programmes. These centres can handle high-capacity needs (Clearview, 2024; Mittra et al., 2024).

The second category, “initiated,” includes health care services with a moderate level of RLT adoption expected to grow in capacity in the short term (within approximately two years). These services (e.g. private urology, radiation oncology practices, or radiologist groups) have basic infrastructure, equipment, and personnel in place and are either starting or planning to start RLT activities soon. They are usually already engaged in practices incorporating radiopharmaceuticals such as iodine-131 for thyroid cancers or radium-223 (Xofigo) for the treatment of patients with castration-resistant prostate cancer and symptomatic bone metastases.

Finally, an “emerging” service refers to health care facilities or centres in the early stages of developing capabilities for theranostic services. These services typically lack the necessary infrastructure, such as dedicated radio-protected chambers, and require significant upgrades to their equipment and workflows to meet regulatory and operational standards (The Health Policy Partnership, 2021). They are often constrained by limited resources, including trained personnel and the necessary funding for upgrading in readiness.

With proper planning, investment and collaboration, RLT services can evolve to deliver cutting-edge cancer therapies, improving outcomes and expanding access to advanced treatments. Establishing a successful theranostic service is a complex and multidisciplinary endeavour. Depending on the existing capabilities, a programme may take one to five years to implement (The Health Policy Partnership, 2021). A theranostic service requires a dedicated team that meets regularly to optimise processes. Industry best practices recommend starting with a pilot programme of 5-10 patients to refine workflows before scaling the service to meet community needs.

Chapter 4. Key findings and areas for possible future work

Six areas for possible future work are presented in this section, as pragmatic actions that NEA member countries can advance to materially improve the security of supply for medical radioisotopes and to enable the commercial deployment of promising new innovations, such as RLT.

4.1 Supply and demand forecasting

Key finding

The NEA continues to publish periodic reports to assess the global supply and demand of medical radioisotopes – in particular Mo-99 and Tc-99m – given the periodic disruptions to the global supply and fluctuating demand. These reports, especially the *Supply of Medical Radioisotopes* series (NEA 2018, 2019), have consistently highlighted vulnerabilities in the financing models, the risks of unplanned outages, as well as the need for structural reform in supply chain management. The NEA's work has helped both governments and industry anticipate possible shortages, optimise the planning of production and develop policy that ensures a stable supply chain. The NEA continues to analyse trends in supply and resilience measures through ongoing engagement with producers, processors and regulators.

Possible areas for future work

The NEA has an important role to play in ongoing supply and demand forecasting, through the continuation of the NEA's periodic assessments of the ongoing supply and demand of Mo-99 and Tc-99m, thereby ensuring market stability and the resilience of the supply chain.

Growing clinical adaptation of Lu-177, Ac-225 and other emerging medical radioisotopes means that NEA forecasting should be expanded to include production capacity, regulatory issues and the likely market dynamics for these novel isotopes if future supply needs are to be met.

4.2 Expert group at the NEA

Key finding

The High-Level Group on the Security of Supply of Medical Radioisotopes (HLG-MR) was established by the NEA in 2009 in response to a severe global shortage of Mo-99 and Tc-99m that occurred in 2007-2008 due to prolonged outages of major production reactors. The HLG-MR was created to develop a co-ordinated strategy for securing the long-term supply of medical radioisotopes. Representatives from member governments, industry and international organisations participating in the HLG-MR helped shape international co-operation and promoted sustainable production models through the group's periodic reports, policy recommendations and consultations with stakeholders. In particular, the HLG-MR recommended that the principle of full-cost recovery (FCR) be applied to medical radioisotopes and sought to align pricing mechanisms with the actual cost of producing these isotopes – thereby reducing dependence on government subsidies and fostering stability driven by the market.

The HLG-MR was granted four mandates, with the fourth and final mandate concluding in 2018, after which the group was formally dissolved. Significant progress had been achieved in response to the 2007-2008 crisis, which included important policy advancements – such as improved co-ordination among Mo-99 producers, greater transparency in reactor scheduling, and stronger regulatory frameworks. Nevertheless, the medical radioisotope supply chain remains vulnerable to disruptions. Continued vigilance and adaptation will be necessary, particularly as demand grows and new isotopes, such as those supporting the expanding field of RLT, begin to play a larger role in nuclear medicine

Possible areas for future work

The NEA Committee for Technical and Economic Studies on Nuclear Energy Development and the Fuel Cycle (NDC) may propose a mandate for a new Expert Group on Medical Radioisotopes (EG-MR) to address ongoing and emerging challenges in the medical radioisotope supply chain. Should the mandate be both finalised and approved by NDC in 2025, the EG-MR could be launched in January 2026.

This group would monitor Mo-99 production and demand trends, assess the pipeline of innovative radioisotopes, provide policy recommendations, and follow up on the HLG-MR's final report recommendations to ensure long-term supply security and resilience.

4.3 International market transparency and monitoring of supply

Key findings

The Nuclear Medicine Europe (NMEU) Security of Supply Working Group (SoS WG) plays a central role in monitoring international markets for medical radioisotopes. NMEU is not limited to European stakeholders; it includes key industry players from around the world, including producers, processors and health care providers. The working group provides real-time updates on reactor operations, outages, and Mo-99 processing capacity, ensuring that stakeholders can respond swiftly to potential disruptions. By gathering and disseminating supply chain data, the NMEU enhances transparency in the medical radioisotope market, helping to mitigate risks associated with production bottlenecks and logistical challenges.

The NMEU focuses on short-term supply chain monitoring, which is complementary to the work of the NEA on longer-term supply and demand forecasting. Through these complementary efforts, both NMEU and the NEA support international co-operation, ensuring that stakeholders – from producers to policymakers – have access to reliable data to anticipate and respond to fluctuations in medical radioisotope supply.

The NEA has participated in NMEU on an ad hoc basis in the past.

Possible areas for future work

NEA member countries could reinforce participation in the NMEU SoS WG to enhance real-time monitoring and transparency in the medical radioisotope supply chain.

Additionally, with the support of its member countries, the NEA could explore opportunities to formalise previously ad hoc collaborations with NMEU. This would help ensure comprehensive short- and long-term assessments of market stability, production sustainability, and infrastructure needs, enabling informed policy decisions and supply chain resilience.

4.4 National policy frameworks

Key finding

The oversight and management of medical radioisotope production and supply is often fragmented across multiple government ministries and agencies, which can hinder policy implementation and co-ordination. In many countries, the responsibilities are divided between the ministries of energy, health, innovation and industry, with no single entity fully overseeing the sector. While ministries of health regulate clinical applications and reimbursement policies, energy ministries oversee policies for the distribution of medical radioisotopes, and industry ministries support technological innovation. This division of responsibility can lead to inefficiencies, delays in decision-making, and challenges in securing funding for production infrastructure, import mechanisms, or emergency supply measures. In some cases, cross-ministerial co-ordination occurs on an ad hoc basis, undermining the ability of governments to enact enabling frameworks and disburse funds and measures efficiently.

Ensuring clear inter-ministerial collaboration is essential for establishing effective funding mechanisms, aligning national production strategies with global supply needs, and implementing policies that support both existing and emerging production technologies. Without a co-ordinated national framework, the sector remains vulnerable to external shocks, underinvestment and inefficiencies that can compromise access to essential medical radioisotopes.

Possible areas for future work

Governments could establish national policy frameworks that bring together all relevant ministries, regulators and industry stakeholders to improve co-ordination in medical radioisotope production and supply. These frameworks would clearly define responsibilities, streamline decision-making, and establish points of contact for external stakeholders to enhance supply security and accelerate the deployment of innovative RLT and other emerging isotopes. The NEA will continue to assist NEA member countries and potential markets to ensure resilient and secure supply chains for the delivery of medical radioisotopes to patients internationally.

4.5 Regulatory efficiency

Key finding

Differences in regulatory requirements across jurisdictions create challenges in the medical radioisotope supply chain, leading to some inefficiencies and delays in isotope production and distribution. One notable example is differences in licensing and approval timelines for radiopharmaceuticals. While some countries have streamlined regulatory pathways for the introduction of new medical radioisotopes, others impose lengthy and complex approval processes. For instance, Lu-177-based therapies received early approval and reimbursement inclusion in Belgium, Canada, Germany and the United States, while in other OECD countries, the integration into national health care systems takes significantly longer.

Additionally, patient discharge standards in RLT vary across countries, reflecting differences in safety regulations and health care protocols. In the United States, patients receiving Lu-177-based therapies are usually treated as outpatients and are allowed to go home the same day with thorough safety information. In Germany, by contrast, due to differing radiation safety rules, the patients may have to stay overnight in a special ward for as much as 48 hours to monitor radiation decay. This naturally impacts the number of patients treated weekly in each country. Thus, outpatient-based centres in the United States may have a higher patient throughput than facilities in Germany, affecting the economics of treatment facilities.

Possible areas for future work

The NEA and its member countries could explore opportunities to strengthen international collaboration to improve regulatory efficiencies in the approval and distribution of medical radioisotopes, ensuring streamlined access to critical treatments.

In particular, there may be opportunities to harmonise regulatory standards for new emerging isotopes such as Lu-177 and Ac-225, facilitating consistent safety protocols, production practices and patient discharge regulations across jurisdictions.

4.6 Enabling innovation

Key finding

RLT represents a significant advancement in precision medicine, offering highly targeted approaches to cancer and other diseases. RLT combines the specificity of molecular targeting with the therapeutic potential of radionuclides, enabling precise destruction of cancer cells while sparing healthy tissue. This innovation has demonstrated strong clinical benefits in conditions such as neuroendocrine tumours and prostate cancer, resulting in the approval of novel anti-cancer drugs. Additionally, new isotopes, including actinium-225 (Ac-225) and copper-67 (Cu-67), show promise for expanding the range of treatable cancers. However, despite these advancements, the adoption of these therapies faces multiple challenges, including limited production capacity and challenges associated with developing and scaling novel isotopes (Financial Times, 2024).

Regulatory hurdles and reimbursement complexities further impede widespread clinical adoption. Differences in national regulatory frameworks create barriers to approval and market entry, delaying patient access to innovative treatments. Additionally, not all health care systems have incorporated RLT and novel isotopes into reimbursement schemes, leading to disparities in availability across regions.

Infrastructure gaps in clinical facilities, including the need for specialised equipment and radiation safety measures, further hinder widespread implementation. Moreover, public and physician awareness of RLT remains limited, affecting early diagnosis and referrals, while a shortage of trained professionals in nuclear medicine and radiopharmacy poses additional barriers to effective adoption.

Possible areas for future work

Through national strategies and international collaborations undertaken under the aegis of the NEA and NMEU, governments and stakeholders could realise enabling conditions for the large-scale development and deployment of novel radioisotopes, to ensure supply will scale to meet demand and improve patient access.

Essential enabling conditions would include, among other considerations, specialised infrastructure and logistics, efficient regulatory pathways and workforce training.

Chapter 5. Conclusions

5.1 Summary of key challenges and opportunities

This publication identified two major challenges:

- ensuring security of supply for medical radioisotopes, including but not limited to Mo-99; and
- enabling the adoption of radioligand therapy (RLT).

The reliable supply of Mo-99, the cornerstone isotope for diagnostics, remains a critical challenge. The production of Mo-99 currently depends on a small number of ageing reactors, many of which face operational inefficiencies and require frequent maintenance. Geopolitical risks, logistical constraints, and the transition from HEU to LEU have further revealed vulnerabilities in the supply chain (NEA, 2019).

Modernising infrastructure and diversifying production methods are essential to address these issues. For example, in Canada, OPG subsidiary Laurentis Energy Partners is working to harvest Mo-99 from OPG's CANDU reactors at its Darlington Nuclear Station, which will make Darlington the first commercial-scale reactor in North America to produce Mo-99. Bruce Power announced in 2024 that it is also exploring production of new medical radioisotopes from the commercial CANDU reactors it currently operates.

Alongside reactor-based initiatives like PALLAS in the Netherlands, technological innovations in accelerator-driven and cyclotron-based production offer promising alternatives. These include the SHINE Technologies work to develop a fusion-driven, non-reactor approach to Mo-99 production, which would eliminate the need for traditional reactors while ensuring a reliable large-scale supply. In parallel, TRIUMF is developing a cyclotron-based method to directly produce Tc-99m from Mo-100 via proton bombardment, which would enable local, on-demand production in hospitals.

Equally important, RLT presents a novel approach to cancer treatment, offering precision-targeted therapies with significant clinical benefits. However, its adoption is hindered by infrastructure gaps, insufficient numbers of trained health care professionals, and inconsistent regulatory frameworks. These challenges limit the availability of RLT, particularly in low-resource settings.

Opportunities to advance RLT include investments in specialised infrastructure, such as treatment facilities with advanced imaging and radiopharmacy capabilities. Expanding training programmes for nuclear medicine professionals and fostering international collaboration on regulatory alignment could aid this process. Moreover, establishing reimbursement mechanisms will promote open access to RLT and ensure its long-term sustainability.

By addressing these challenges and leveraging these opportunities, nuclear medicine can achieve a secure supply of medical radioisotopes while accelerating the adoption of RLT.

5.2 Ensuring the future of medical radioisotope supply

If well maintained and well scheduled, current irradiators and processors involved in producing Mo-99 will suffice to respond to limited periods of unplanned outage in the near to medium term. However, concerns remain about the relative age of many of the facilities used in producing Mo-99 and other radiological isotopes. While medical radioisotopes internationally now benefit from a decade of investment undertaken by many countries, the extended time needed to replace facilities and introduce new technologies continues to be a significant challenge. The advent of additional radioisotopes with novel applications in areas such as diagnosis and therapeutics will only add significance to this challenge moving forward.

5.3 The role of policy in strengthening the radioisotope supply chain

The advent of RLT technologies and novel radioisotopes, including Lu-177 and Ac-225, presents challenges and opportunities for health systems. Adapting to these advancements requires strategic planning, infrastructure development, and a collaborative approach to successfully integrating these innovative therapies. This includes, inter alia, specialised training and education; regulatory compliance; patient access; cost and reimbursement strategies; hospital investments in specialised radiation therapy delivery systems; and clinical pathways and guidelines.

5.4 Next steps

The proposal to re-establish a medical radioisotope programme stems from the NEA International Workshops on Medical Radioisotopes Supply held in 2023 and 2024. The turnout and engaged discussion during the events indicate continued interest from governments and the private sector for enhanced co-operation in the field. Participants displayed general consensus on the benefits of moving beyond surveys on the state-of-play for Mo-99 to technical work in new areas such as market forecasting for emerging radioisotopes (e.g. Lu-177 for current therapeutic applications and Ac-225 in early-stage development).

Understanding the breadth and timeline for the global integration of novel radioisotopes is crucial as the demand for advanced therapeutic and diagnostic tools continues to rise. The NEA aims to undertake, in collaboration with the Health Division of the OECD Directorate for Employment, Labour and Social Affairs, as well as with selected industry stakeholders, a broad study on the state, potential and challenges of RLT for cancer treatment. Mindful of the differences between national health care systems, the study will aim to develop policy recommendations for coherent frameworks for RLT in OECD member countries. The study on market forecasting for emerging radioisotopes will include, but will not be restricted to, the following elements:

- An overview of the current state of RLT in OECD member countries regarding scientific research, therapeutic results, and patient benefits. This would include taking stock of a dynamic innovation pipeline and identifying the most promising radioisotopes, timelines and uncertainties.
- An assessment of the medium-term market potential in the next ten years for RLT under different assumptions for baseline anticipated growth, low growth and high growth in the context of the required investments in supply chain and hospital infrastructure. This would involve scenario modelling and simulation tools to project the demand (including an analysis of the epidemiological data related to various cancer types), supply, system costs for the supply chain, and benefits of RLT.
- The identification of the main challenges to a broad and sustained uptake of RLT (e.g. health care infrastructure readiness, institutional lags, insufficient workforce

preparation, administrative hurdles, and, particularly, bottlenecks regarding the supply, processing and logistics of medical radioisotopes.)

- The development of policy recommendations, based on stakeholder and medical radioisotope community engagement, addressed to decision makers in the health care systems of OECD countries, regarding appropriate frameworks for RLT in diagnosing and treating cancers, thereby maximising patient benefits cost-effectively.

Work on the study will be completed in Q1 2026. Activities will be conducted in consultation with experts from the medical community, industry and government. The proposed work includes literature reviews, face-to-face interviews and questionnaires. Given RLT's relatively new application in medical treatment, challenges include a lack of awareness about the technology, an absence of harmonisation regarding education, training, and benchmarking of best practices, a lack of consistent criteria for approval and reimbursement, and, crucially, ensuring a sustainable supply for all medically relevant radioisotopes. The need for health care systems to be aware and prepared for a wider-spread adoption of RLT is all the more urgent, as RLT requires specific readiness for health care infrastructures, workforce, regulations, and administrative and financial arrangements.

Proposed activities for the NEA work on medical radioisotopes include holding an annual international workshop, undertaking additional studies, and creating the Expert Group on Medical Radioisotopes under the NEA's Committee for Technical and Economic Studies on Nuclear Energy Development and the Fuel Cycle (NDC). Together, these activities will help inform member countries' policy decisions needed to accelerate and expand access to these potentially life-saving nuclear medicine technologies.

Glossary

Alpha emitter	A radioactive substance that releases alpha particles (helium nuclei) during its decay process. Alpha particles have high energy but travel only short distances (<100µm).
Beta emitter	A radioactive substance that releases beta particles (electrons or positrons) during its decay process. Beta particles have moderate energy and can travel a few millimetres in biological tissue.
Carrier	A stable (non-radioactive) isotope of the same element that is present alongside the desired radioactive isotope, affecting its purity and specific activity.
Decay	The process by which an unstable atomic nucleus transforms into a more stable state by emitting radiation (such as alpha particles, beta particles or gamma rays). The rate of decay depends on the element and is characterised by its half-life.
Diagnostic isotopes	Medical radioisotopes that emit gamma rays, which can penetrate human tissue and be detected by imaging devices (e.g. PET, SPECT) to visualise physiological and pathological processes. An example is technetium-99m (Tc-99m).
Fission targets	A material that, when irradiated with a neutron beam in a nuclear reactor undergoes nuclear reactions to produce radioactive isotopes.
Half-life	The time required for a radioactive sample to decay to half of its original radioactivity due to nuclear decay.
Just-in-time	Manufacturing and supply approach where the final product is produced and delivered only as needed.
Ligand	A molecule that binds with high affinity to a specific biological compound usually found on the cell surface.
Molecular targets	Receptors or proteins found on the cell surface (or within the cell) that serve as attachment points for molecules, which can trigger biological responses, alter cell function, or be used for imaging and therapeutic purposes.
Molybdenum-99 (Mo-99)	Mo-99 is the most commonly used radioisotope in nuclear medicine diagnostic scans. Mo-99 has a half-life of 66 hours, that is, its radioactivity decreases by half in 66 hours.
Refuelling	The process of replacing or replenishing nuclear fuel in a reactor to sustain the fission reaction.

Technetium-99m (Tc-99m)	Tc-99m is obtained from radioactive decay of its parent isotope, Mo-99 and has a half-life of just 6 hours.
Theranostics	A medical approach that combines diagnostic imaging and targeted therapy using radiopharmaceuticals to personalise treatment and improve patient outcomes.
Therapeutic isotopes	Radioisotopes that emit alpha or beta particles, which have higher energy but short penetration depths. These properties allow them to damage cancer cells when specifically targeted to the tumour site. Examples include lutetium-177 (beta emitter) and actinium-225 (alpha emitter).

References

ARPANSA (n.d.a), “Alpha particles”, website article, available at: www.arpansa.gov.au/understanding-radiation/what-is-radiation/ionising-radiation/alpha-particles (accessed 26 March 2025).

ARPANSA (n.d.b), “Beta particles”, website article ,available at: www.arpansa.gov.au/understanding-radiation/what-is-radiation/ionising-radiation/beta-particles (accessed 26 March 2025).

ARTMS (2020), "Health Canada approves cyclotron produced technetium-99m utilizing the ARTMS QIS and solid targets", available at: https://artms.ca/health-canada-approves-cyclotron-produced-technetium-99m-utilizing-the-artms-qis-and-solid-targets/ (accessed 20 February 2025).

Bruce Power (2024), "Bruce Power to explore production of new medical isotopes, expansion of Isotope Production System", available at: https://www.brucepower.com/2024/11/01/bruce-power-to-explore-production-of-new-medical-isotopes-expansion-of-isotope-production-system/ (accessed 20 January 2025).

Bruce Power (2022), "Isotope Production System begins commercial production of cancer-fighting lutetium-177", *Bruce Power*, 24 October, available at: www.brucepower.com/2022/10/24/isotope-production-system-begins-commercial-production-of-cancer-fighting-lutetium-177/ (accessed 26 February 2025).

Buglione, F. (2016), "Evaluation of radioisotope production capabilities in MYRRHA", Master's thesis for Politecnico di Torino, available at: https://roma.sckcen.be/ws/portalfiles/portal/4565799/Evaluation_of_Radioisotope_Production_Capabilities_in_MYRRHA.pdf (accessed 11 August 2025).

BWXT (2023), "BWXT Medical and Laurentis complete key Tc-99m generator program milestone", available at: www.bwxt.com/news (accessed 20 January 2025).

Clearview (2024), "Radioligand therapy: Seizing the opportunity", available at: https://clearviewhcp.com/wp-content/uploads/2024/03/Radioligand-Therapy-Seizing-The-Opportunity.pdf.

CNEA (2024), "The National Atomic Energy Commission is carrying out the construction of the RA-10 Argentine Multipurpose Nuclear Reactor, located at the Ezeiza Atomic Centre", available at: www.argentina.gob.ar/ra-10-multipurpose-reactor (accessed 20 January 2025).

Cutler, C.S. and S.W. Schwarz (2014), "Diversification in the supply chain of 99Mo ensures a future for 99mTc", *Journal of Nuclear Medicine*, 55(7), pp. 1208–1213, doi:10.2967/jnumed.113.131953.

Czernin, J. and J. Calais (2022), "How many theranostics centers will we beed in the United States?", *Journal of Nuclear Medicine*, 63(6), pp. 805–806. doi:10.2967/jnumed.122.264144.

Deshayes, E. et al. (2017), "Radium 223 dichloride for prostate cancer treatment", *Drug Design, Development and Therapy*, 11, pp. 2643–2651, doi:10.2147/DDDT.S122417.

EC (2025), *State Aid Procedures – European Commission*, available at: https://competition-policy.ec.europa.eu/state-aid/procedures_en (accessed 4 February 2025).

Fahey, F.H. and F.D. Grant (2021) "Celebrating eighty years of radionuclide therapy and the work of Saul Hertz", *Journal of Applied Clinical Medical Physics*, 22(1), pp. 4–10, doi:10.1002/acm2.13175.

Farolfi, A. et al. (2024), "Differences and common ground in 177Lu-PSMA radioligand therapy practice patterns: International survey of 95 theranostic centers", *Journal of Nuclear Medicine*, 65(3), pp. 438–445, doi:10.2967/jnumed.123.266391.

Ferrucci, B., et al (2022), "Future development of global molybdenum-99 production and saving of atmospheric radioxenon emissions by using nuclear fusion-based approaches", *Journal of Environmental Radioactivity*, 255 (12), doi: 10.1016/j.jenvrad.2022.107049.

Financial Times (2024), "The hunt for a rare nuclear isotope that could redefine cancer care", *Financial Times*, 28 July, available at: www.ft.com/content/6ce668bc-4180-4e84-9feb-f25ac0e83f6f (accessed 11 August 2025).

George, S.C. et al. (2024), "A single-institution experience with 177Lu RPT workflow improvements and qualifying the SPECT/CT imaging for dosimetry", *Frontiers in Oncology*, 14, doi:10.3389/fonc.2024.1331266.

Government of Canada (2023), "Government strengthens Canada’s leadership role in medical isotope production", available at: www.canada.ca/en/innovation-science-economic-development/news/2023/06/government-strengthens-canadas-leadership-role-in-medical-isotope-production.html (accessed 20 January 2025).

Halfdanarson, T.R. et al. (2023), "Monitoring and surveillance of patients with gastroenteropancreatic neuroendocrine tumors undergoing radioligand therapy", *Cancers*, 15(19), p. 4836, doi:10.3390/cancers15194836.

Herrmann, K. et al. (2022), "Joint EANM, SNMMI and IAEA enabling guide: how to set up a theranostics centre", *European Journal of Nuclear Medicine and Molecular Imaging*, 49(7), pp. 2300–2309, doi:10.1007/s00259-022-05785-x.

Horwitz, E.P. et al. (2005), "A process for the separation of 177Lu from neutron irradiated 176Yb targets", *Applied Radiation and Isotopes*, 63(1), pp. 23–36, doi:10.1016/j.apradiso.2005.02.005.

IAEA (2025), "IMAGINE – SPECT Scanners (per 1 mil)", online database available at https://humanhealth.iaea.org/HHW/DBStatistics/IMAGINEMaps5.html (accessed 26 March 2025).

IRE (2023), "IRE moves to 100% LEU (low-enriched uranium) conversion", available at: www.ire.eu/en/media-room/ire-moves-to-100-leu-low-enriched-uranium-conversion.

Jang, A. et al. (2023), "Targeted alpha-particle therapy: A review of current trials", *International Journal of Molecular Sciences*, 24(14), p. 11626, doi:10.3390/ijms241411626.

Jeelani, S. et al. (2014), "Theranostics: A treasured tailor for tomorrow", *Journal of Pharmacy & Bioallied Sciences*, 6(Suppl 1), pp. S6-8, doi:10.4103/0975-7406.137249.

"JHR services" (2025), *JHR*, available at: https://jhrreactor.com/en/jhr-services/ (accessed 18 February 2025).

Kazemi-Jahromi, M. et al. (2024), "Personalized dosimetry assessment of [177Lu]Lu-PSMA-617 radioligand therapy in the management of metastatic castration-resistant prostate cancer", *International Journal of Radiation Biology*, 100(11), pp. 1551–1559, doi:10.1080/09553002.2024.2404448.

KINECTRICS (2024), "Global supply chain for Lu-177 based cancer treatments strengthened by Kinectrics' start of Yb-176 production, reducing reliance on Russia", available at: www.kinectrics.com/media-center/news-releases/global-supply-chain-for-lu-177-based-cancer-treatments-strengthened-by-kinectrics-start-of-yb-176-production-reducing-reliance-on-russia (accessed 24 January 2025).

Koleska, M. et al. (2011), "Completed conversion of the LVR-15 research reactor", in. *ERTR 2011: 33, International Meeting on reduced Enrichment for Research and Test Reactors*, Santiago (Chile), available at: www.osti.gov/etdeweb/biblio/22008670 (accessed 20 January 2025).

Laurentis (2023), "Darlington Nuclear now capable of producing life-saving medical isotopes", Laurentis Energy Partners, available at: https://laurentisenergy.com/releases/darlington-nuclear-now-capable-of-producing-life-saving-medical-isotopes/ (accessed 26 February 2025).

LLP, D.B.R. (2024), "Radioligand therapies market to show immense growth by 2034, predicts DelveInsight", available at: www.prnewswire.com/news-releases/radioligand-therapies-market-to-show-immense-growth-by-2034-predicts-delveinsight--key-companies---curium-us-eli-lilly-point-biopharma-fusion-astrazeneca-302191571.html (accessed 18 March 2025).

Malandrino, P. et al. (2024), "Radioligand Therapy in Patients with Lung Neuroendocrine Tumors: A Systematic Review on Efficacy and Safety", *Seminars in Nuclear Medicine*, 54(4), pp. 570–580, doi:10.1053/j.semnuclmed.2024.05.001.

Migdal, M. and T. Krok (2014), "Brief history of MARIA conversion from HEU to LEU", in. *RERTR-2014 International Meeting*, Vienna, Austria.

Mittra, E.S. et al. (2024), "Establishing a robust radioligand therapy program: A practical approach for North American centers", *Cancer Medicine*, 13(3), p. e6780, doi:10.1002/cam4.6780.

MSKCC (2022), "About Your Pluvicto™ Treatment", Memorial Sloan Kettering Cancer Center website, available at: www.mskcc.org/cancer-care/patient-education/about-your-pluvicto-treatment (accessed 23 January 2025).

MYRRHA (2024), "Construction of world first MYRRHA underway", SCK CEN website, available at: www.sckcen.be/en/news/construction-world-first-myrrha-underway (accessed 20 January 2025).

Naoyuki, T. et al. (2023), "Domestic Production of Mo-99 and Ac-225 using commercial PWR and fast experimental reactor JOYO in Japan", *11th International Conference on Isotopes(11ICI)*, TCU Place, Saskatoon, Saskatchewan, Canada, available at: https://harvest.usask.ca/server/api/core/bitstreams/5bf74a96-c610-4ee0-8fe3-2c56cb6a6a96/content (accessed 20 January 2025).

National Institute for Health and Clinical Excellence (2009), "Guide to the single technology appraisal process", NHS, available at: www.nice.org.uk/media/default/about/what-we-do/nice-guidance/nice-technology-appraisals/guide-to-the-single-technology-appraisal-process.pdf (accessed 27 March 2025).

NCBJ (2024), "The government adopted a resolution on the project to modernize the MARIA nuclear research reactor", available at: www.ncbj.gov.pl/en/news/government-adopted-resolution-project-modernize-maria-nuclear-research-reactor (accessed 20 January 2025).

NEA (2023), *The Security of Supply of Medical Radioisotopes: Demand and Capacity Projections for 99Mo/99mTc for the 2023-2027 Period*, OECD Publishing, Paris, available at: www.oecd-nea.org/radioisotopes23-27 (accessed 20 January 2025).

NEA (2019), *The Supply of Medical Radioisotopes: An Economic Diagnosis and Possible Solutions,* OECD Publishing, Paris, available at: www.oecd-nea.org/jcms/pl_19862 (accessed 31 March 2025).

NEA (2018), *The Supply of Medical Radioisotopes: 2018 Medical Isotope Demand and Capacity Projection for the 2018-2023 Period*, OECD Publishing, available at: www.oecd-nea.org/jcms/pl_19862 (accessed 31 March 2025).

NORTHSTAR (2024), "NorthStar Medical Radioisotopes, LLC Enters Clinical Supply Agreement to Support Clarity Cu-67 SAR-bisPSMA Pivotal Trials", available at: www.northstarnm.com/northstar-medical-radioisotopes-llc-enters-clinical-supply-agreement-to-support-clarity-cu-67-sar-bispsma-pivotal-trials/ (accessed 27 January 2025).

NORTHSTAR (2023), "NorthStar Medical Radioisotopes sets new industry precedent in advancing non-uranium based commercial production of molybdenum-99 (Mo-99)", NorthStar Medical Radioisotopes website, available at: www.northstarnm.com/northstar-medical-radioisotopes-sets-new-industry-precedent-in-advancing-non-uranium-based-commercial-production-of-molybdenum-99-mo-99/ (accessed 28 March 2025).

Norwegian Ministry of Foreign Affairs (2018), "Report on Radioisotope Production By Working Group 3 Chaired by Korea", *The 3rd International Symposium on HEU minimization*, Oslo, Norway, available at: www.stimson.org/wp-content/uploads/2021/01/WG3-OSLO2.pdf (accessed 20 January 2025).

NRG (2024), "Green light from the European Commission for the construction of the PALLAS-reactor", NRG website, available at: www.nrg.eu/en/news/green-light-from-the-european-commission-for-the-construction-of-the-pallas-reactor (accessed 20 January 2025).

Oliver Sartor (2023), "ESMO 2023: PSMAfore Phase 3 Trial of [177Lu]Lu-PSMA-617 in taxane-naive patients with metastatic castration resistant prostate cancer", available at: www.urotoday.com/conference-highlights/esmo-2023/esmo-2023-prostate-cancer/147587-esmo-2023-psmafore-phase-3-trial-of-177lu-lu-psma-617-in-taxane-naive-patients-with-mcrpc.html (accessed 18 March 2025).

Oncology Nurse Advisor (2024), "Theranostic treatment success achieved through tight care team coordination", website available at: www.oncologynurseadvisor.com/features/theranostics-treatment-team/ (accessed 19 March 2025).

OPG (2025a), "Nuclear isotopes", website available at: www.opg.com/projects-services/projects/nuclear/nuclear-isotopes/ (accessed 26 February 2025).

OPG (2025b), "Darlington Nuclear's newly refurbished Unit 1 now producing life-saving Cobalt-60 isotopes", website available at: www.opg.com/news-resources/newsroom/our-stories/story/darlington-nuclears-newly-refurbished-unit-1-now-producing-life-saving-cobalt-60-isotopes (accessed 26 February 2025).

SCK CEN (2023), "BR2 uses low-enriched uranium as fuel", SCK CEN website, available at: www.sckcen.be/en/news/br2-uses-low-enriched-uranium-fuel (accessed 20 January 2025).

Sgouros, G. et al. (2020), "Radiopharmaceutical therapy in cancer: clinical advances and challenges", *Nature Reviews Drug Discovery*, 19(9), pp. 589–608, doi:10.1038/s41573-020-0073-9.

SHINE Technologies (2024), "SHINE Technologies awarded $32M by DOE/NNSA to support production of molybdenum-99", available at: www.prnewswire.com/news-releases/shine-technologies-awarded-32m-by-doennsa-to-support-production-of-molybdenum-99-302198781.html (accessed 20 January 2025).

SHINE Technologies (2022), "The science behind n.c.a. Lu-177 therapy: A breakthrough in targeted cancer treatment", available at: www.shinefusion.com/blog/science-behind-lu-177-therapy-targeted-cancer-treatment (accessed 24 January 2025).

Singh, S. et al. (2024), "[177Lu]Lu-DOTA-TATE plus long-acting octreotide versus high-dose long-acting octreotide for the treatment of newly diagnosed, advanced grade 2–3, well-differentiated, gastroenteropancreatic neuroendocrine tumours (NETTER-2): an open-label, randomised, phase 3 study", *The Lancet*, 403(10446), pp. 2807–2817, doi:10.1016/S0140-6736(24)00701-3.

Song, H. et al. (2024), "Same-day post-therapy imaging with a new generation whole-body digital SPECT/CT in assessing treatment response to [177Lu]Lu-PSMA-617 in metastatic castration-resistant prostate cancer", *European Journal of Nuclear Medicine and Molecular Imaging*, 51(9), pp. 2784–2793, doi:10.1007/s00259-024-06718-6.

The Health Policy Partnership (2021), "Health system readiness for radioligand therapy in the US", working paper.

Uijen, M.J.M. et al. (2021), "PSMA radioligand therapy for solid tumors other than prostate cancer: background, opportunities, challenges, and first clinical reports", *European Journal of Nuclear Medicine and Molecular Imaging*, 48(13), pp. 4350–4368, doi:10.1007/s00259-021-05433-w.

UKB (2025), "Lutetium-177-PSMA therapy", website available at: www.ukbonn.de/en/nuclear-medicine/our-department/radionuclide-therapy/lu-177-psma-therapie/ (accessed 23 January 2025).

World Cancer Research Fund (2022), "Prostate cancer statistics", website available at: www.wcrf.org/preventing-cancer/cancer-statistics/prostate-cancer-statistics/ (accessed 18 March 2025).

World Nuclear News (2019), "Shine to use 'novel' method for lutetium production, available at: https://world-nuclear-news.org/articles/shine-to-use-novel-method-for-lutetium-production (Accessed: 24 January 2025).

Zhang, J. et al. (2020), "The efficacy of 177Lu-DOTATATE peptide receptor radionuclide therapy (PRRT) in patients with metastatic neuroendocrine tumours: a systematic review and meta-analysis", *Journal of Cancer Research and Clinical Oncology*, 146(6), pp. 1533–1543, doi:10.1007/s00432-020-03181-2.

Zuo, D. et al. (2024), "Astatine-211 and actinium-225: two promising nuclides in targeted alpha therapy", *Acta Biochimica Et Biophysica Sinica* [Preprint], doi:10.3724/abbs.2024206.

NEA PUBLICATIONS AND INFORMATION

The full catalogue of publications is available online at www.oecd-nea.org/pub.

In addition to basic information on the Agency and its work programme, the NEA website offers free downloads of hundreds of technical and policy-oriented reports. The professional journal of the Agency, *NEA News* – featuring articles on the latest nuclear energy issues – is available online at www.oecd-nea.org/nea-news.

An NEA monthly electronic bulletin is distributed free of charge to subscribers, providing updates of new results, events and publications. Sign up at www.oecd-nea.org/tools/mailing list/register.

Visit us on LinkedIn at www.linkedin.com/company/oecd-nuclear-energy-agency or follow us on X (formerly known as Twitter) @OECD_NEA.

Current Trends in the Supply and Utilisation of Medical Radioisotopes

Medical radioisotopes support millions of diagnostic and therapeutic procedures annually. As such, they serve as essential components of modern health care. This report provides extensive coverage of both traditional radioisotope supply challenges for molybdenum-99 and the developing field of radioligand therapy (RLT) and theranostics.

Drawing on more than 15 years of experience to analyse production capacity, innovation pipelines, regulatory frameworks and health care system readiness, this report offers strategic policy recommendations to guarantee long-term sustainable access to established and new radioisotopes for all stakeholders. It will be particularly useful to government officials, health care professionals and industry leaders working to establish resilient supply chains and patient-centred nuclear medicine innovation. It provides a forward-looking view of how policy and investment choices today can shape the future landscape of nuclear medicine.

Nuclear Energy Agency (NEA)
46, quai Alphonse Le Gallo
92100 Boulogne-Billancourt, France
nea@oecd-nea.org **www.oecd-nea.org**

NEA No. 7743